Space Exploration & Aerospace Engineering

Part XI

Space Observatories

Above the atmosphere, telescopes see the whole electromagnetic spectrum without blur. Part XI covers the engineering of space telescopes and the great observatories, from Hubble and Webb to the X-ray, gamma-ray, gravitational-wave and survey missions of today and tomorrow.



Chapters 38–39 · pages 694–734 of the complete manual
First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

Space Exploration & Aerospace Engineering

A complete, illustrated guide to the space ecosystem — from stratospheric balloons to interstellar probes — explaining how every major class of space system works, how it is engineered, and where artificial intelligence is changing it.

First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

Typography and production

Set in Source Serif 4, Inter and JetBrains Mono. Equations typeset with MathJax. Produced from Markdown sources with an automated pipeline (pandoc, Chromium, pypdf); the sources and build scripts accompany the PDF.

Contents of Part XI

PART XI · SPACE OBSERVATORIES

38 Space Telescope Engineering 695

38.1 Why Go to Space? 696
38.2 Optics: Collecting and Focusing Light 698
38.3 Detectors 703
38.4 Keeping Cold and Clean 706
38.5 Pointing and Stability 708
38.6 Where to Put a Telescope 710
38.7 Building, Testing and Servicing 711
38.8 From Photons to Archives 712

39 Great Observatories and Deep-Space Telescopes 715

39.1 Why Observe from Space 716
39.2 NASA's Great Observatories 718
39.3 The James Webb Space Telescope 721
39.4 Reading the Distant Universe: Redshift, Time and Distance 724
39.5 Surveys and Exoplanet Hunters 726
39.6 High-Energy and Time-Domain Astronomy 730
39.7 Microwaves, Radio and Gravitational Waves 731
39.8 The Fleet in 2026 and the Road Ahead 731

This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example "Chapter 24") and to the appendices refer to the complete book.



PART XI

Space Observatories

Above the atmosphere, telescopes see the whole electromagnetic spectrum without blur. Part XI covers the engineering of space telescopes and the great observatories, from Hubble and Webb to the X-ray, gamma-ray, gravitational-wave and survey missions of today and tomorrow.

38 Space Telescope Engineering

39 Great Observatories and Deep-Space Telescopes

Image: NASA/STScI (Webb) (NASA ESA CSA STScI)

38 CHAPTER Space Telescope Engineering

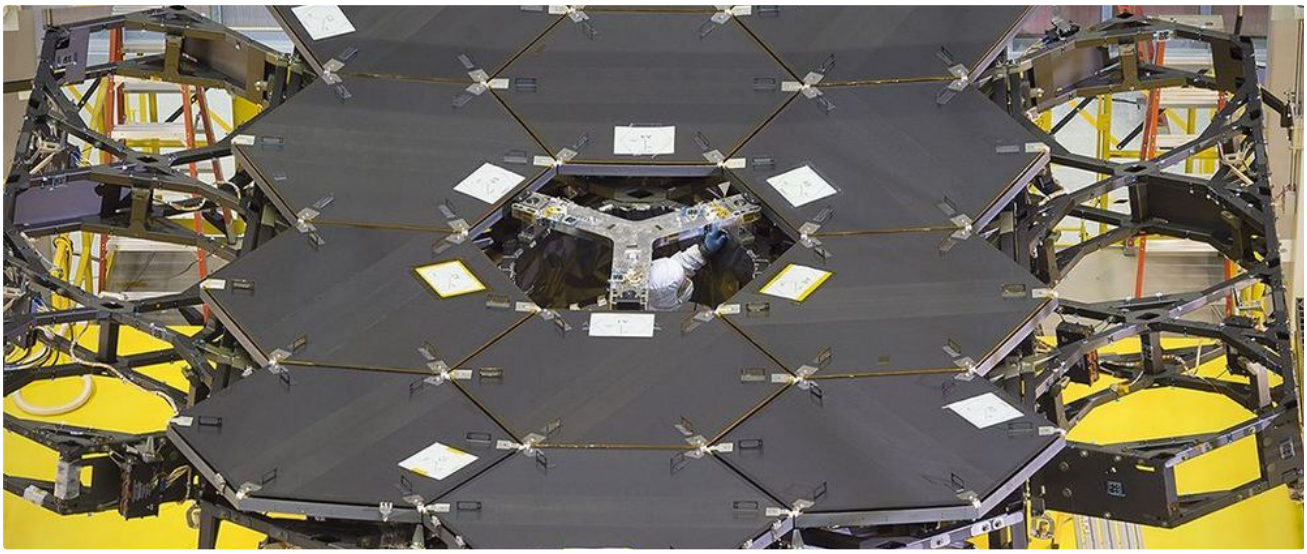


Image: NASA/GSFC

A space telescope is a precision optical instrument that must survive launch, unfold or cool itself in orbit, hold its aim to a few thousandths of an arcsecond and send home data from as far as 1.5 million kilometres away. This chapter explains why astronomers go to space at all — atmospheric absorption, blurring and glow — and then the engineering that follows: diffraction-limited optics, segmented and grazing-incidence mirrors, detectors from CCDs to microcalorimeters, sunshields and cryocoolers, fine guidance and jitter control, the choice of orbit, servicing and testing, and the data chain from downlink to public archive. Worked examples size the resolution, photon budget and thermal shielding of real observatories.

IN THIS CHAPTER

38.1 Why Go to Space?
38.2 Optics: Collecting and Focusing Light
38.3 Detectors
38.4 Keeping Cold and Clean
38.5 Pointing and Stability
38.6 Where to Put a Telescope
38.7 Building, Testing and Servicing
38.8 From Photons to Archives

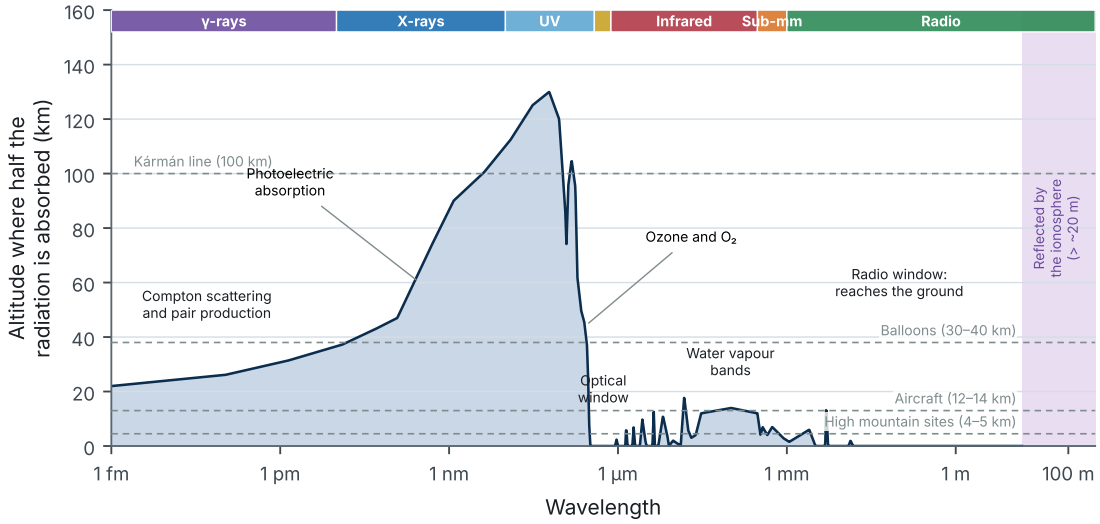
The first view of the Sun from above the ozone layer came in October 1946, when a spectrograph on a captured V-2 rocket recorded its ultraviolet spectrum. Eighty years later, the James Webb Space Telescope (JWST) observes galaxies as they were less than 300 million years after the Big Bang. Its 6.5 m segmented mirror is held below 50 K behind a sunshield the size of a tennis court. The 36-year-old Hubble Space Telescope still works in low Earth orbit. The Nancy Grace Roman Space Telescope, launched on 30 August 2026, is on its way to the Sun–Earth L2 point.

Every one of these observatories is a compromise between physics and engineering. A larger mirror collects more light and sees finer detail, but must fit inside a rocket fairing. A colder telescope sees fainter infrared sources, but needs sunshields or refrigerators. A more distant orbit is thermally quiet, but cannot be visited for repairs and has a weaker radio link. This chapter explains these trade-offs. Optical payloads and radiometry in general are introduced in Chapter 12; the history and science of individual observatories are covered in Chapter 39, and in-space servicing and assembly in Chapter 40.

38.1 Why Go to Space?

38.1.1 The Atmosphere as a Filter

Earth's atmosphere is transparent in only two broad windows: visible light with a few near-infrared bands, and radio waves from about 1 cm to 20 m (Figure 38.1). Gamma rays and hard X-rays are scattered and absorbed by nuclei and electrons, so that even balloons at 35–40 km lose much of the signal. Soft X-rays and extreme ultraviolet light are absorbed by nitrogen and oxygen at 80–150 km, which is why X-ray astronomy beyond the Sun began on a sounding rocket in 1962 (Chapter 9). Ozone absorbs ultraviolet light below about 300 nm. Water vapour and carbon dioxide absorb most of the infrared, leaving narrow windows at high, dry sites. The ionosphere reflects radio waves longer than about 20 m. Table 38.1 lists the lowest platform from which each band can be observed.



Schematic: approximate half-absorption altitudes after Giacconi et al. (1968) and standard texts; X/γ values from air attenuation coefficients and the standard atmosphere.

FIGURE 38.1 Altitude at which half of the incoming radiation from space has been absorbed, as a function of wavelength. Only visible light and radio waves reach the ground; X-ray, gamma-ray, ultraviolet and most infrared astronomy must be done from balloons, rockets or spacecraft.

38.1.2 Seeing, Diffraction and Sky Brightness

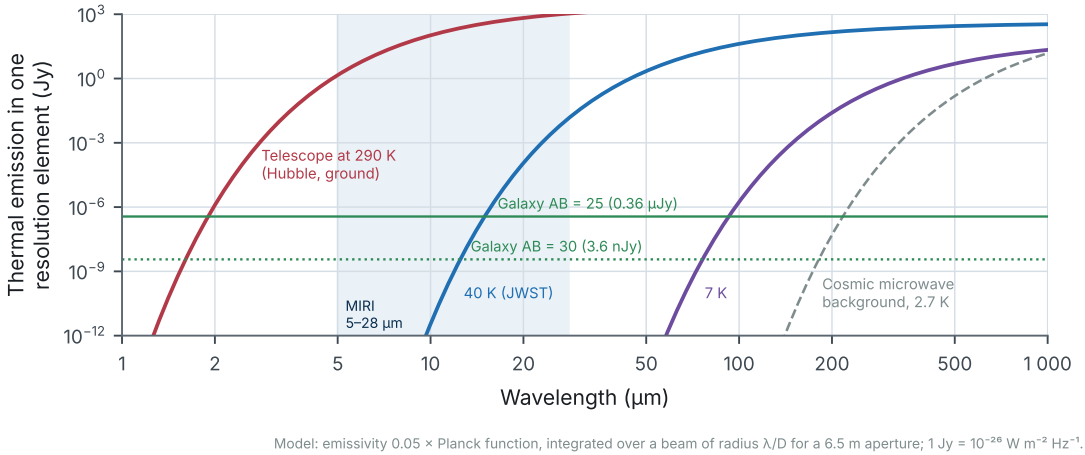
Even where the air is transparent, it is not still. Turbulent cells of air at slightly different temperatures act as moving lenses and smear a star into a flickering blob — astronomical **seeing**. At the best mountain sites the blur is about 0.4–1 arcsecond (″; 1″ = 4.85 µrad), whatever the telescope's size. A 2.4 m telescope in space is limited instead by **diffraction** to about 0.05″ in

visible light. Ground telescopes now partly correct seeing with **adaptive optics**. But the correction works best in the infrared, over small fields near a bright guide star. It never matches the stability of a space image, which can stay the same for years.

The atmosphere also **glows**: emission lines from hydroxyl radicals near 90 km make the near-infrared night sky bright. In space the darkest background is **zodiacal light**, sunlight scattered and re-emitted by interplanetary dust. Its visible brightness is about 22–23 magnitudes per square arcsecond — somewhat darker than the darkest ground sites. In the near infrared it is hundreds of times darker than the sky seen from the ground. It is this background that sets the limiting magnitude of Hubble and JWST.

38.1.3 The Infrared Problem: Telescopes That Glow

Everything warm emits infrared radiation. A telescope at room temperature radiates like a blackbody peaking near 10 µm, multiplied by its mirrors' emissivity of a few per cent. Figure 38.2 shows the consequence. At 10 µm a 290 K telescope puts a few hundred million times more power into each resolution element than a galaxy of AB magnitude 25. The only cure is cold. At 40 K, JWST's own emission is fainter than even a 30th-magnitude galaxy below about 12 µm and dominates its background only at the longest wavelengths of its Mid-Infrared Instrument (MIRI). Herschel's mirror, at about 85 K, was limited by its own glow, and future far-infrared missions aim for a few kelvin. This single fact drives the sunshields, orbits and cryogenics described later.



Model: emissivity 0.05 × Planck function, integrated over a beam of radius λ/D for a 6.5 m aperture; 1 Jy = 10^{-26} W m^{-2} Hz^{-1}.

FIGURE 38.2 Thermal emission from a telescope (mirror emissivity 0.05) at three temperatures, compared with faint galaxies of AB magnitude 25 and 30. Cooling to 40 K removes the problem below about 12–15 µm; only a few-kelvin telescope is dark in the far infrared.

TABLE 38.1 Where each band of the electromagnetic spectrum can be observed from.

Band	Wavelength / energy	Main absorber	Lowest useful platform	Space telescopes
Gamma rays	> 100 keV	Nuclei, electrons	Balloon (partly), satellite	Fermi, Swift BAT
X-rays	0.1–100 keV	Photoelectric absorption	Rocket, satellite	Chandra, XMM-Newton, XRISM
Ultraviolet	10–320 nm	O_2, O_3	Rocket, satellite	Hubble, GALEX
Visible	0.32–0.9 µm	(none)	Ground (blurred)	Hubble, Gaia, Euclid
Near/mid infrared	0.9–30 µm	H_2O, CO_2	Mountain (windows only)	JWST, Spitzer, Roman

Band	Wavelength / energy	Main absorber	Lowest useful platform	Space telescopes
Far infrared	30–300 μm	H_2O	Aircraft, balloon	Herschel, planned PRIMA
Sub-mm and radio	0.3 mm–20 m	H_2O (sub-mm)	Ground, high dry sites	Planck (microwave background)

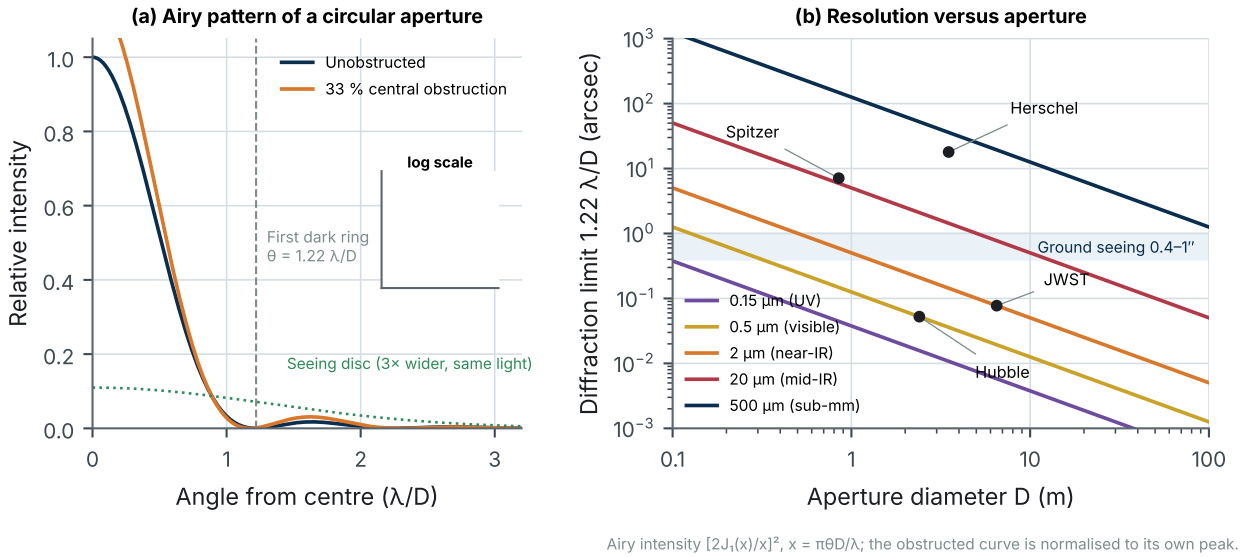
38.2 Optics: Collecting and Focusing Light

38.2.1 The Diffraction Limit

A telescope **collects** light in proportion to its area and **resolves** detail in proportion to its diameter. Because light is a wave, a circular aperture of diameter D focuses a point source into an **Airy pattern** — a bright core surrounded by faint rings — whose first dark ring lies at

$$\theta = 1.22\,\frac{\lambda}{D}$$

where λ is the wavelength (Figure 38.3). Two stars closer than about this angle merge into one (the **Rayleigh criterion**). The core holds 84 % of the light; a central obstruction, such as a Cassegrain secondary mirror, moves light into the rings, which matters for high-contrast imaging. To exploit the diffraction limit, the detector must have at least two pixels across λ/D (**Nyquist sampling**). Hubble's cameras, with 40–50 mas (milliarcsecond) pixels, are undersampled about twofold in the visible (Nyquist at 0.5 μm needs 21 mas). They recover detail by combining several offset ("dithered") exposures.



Airy intensity $[2J_1(x)/x]^2$, $x = \pi\theta D/\lambda$; the obstructed curve is normalised to its own peak.

FIGURE 38.3 (a) The Airy pattern of a circular aperture and the effect of a central obstruction, compared with a seeing-limited image of the same total flux. (b) Diffraction-limited resolution versus aperture at five wavelengths; ground-based seeing is shown for comparison.

EXAMPLE: HOW SHARP IS A 6.5 M TELESCOPE?

JWST has an effective diameter of 6.5 m. What is its resolution at 2 µm, and how does it compare with Hubble and with the ground?

1. Diffraction limit: $\theta = 1.22\times2\times10^{-6}/6.5$ = 3.75 × 10^{-7} rad. Multiplying by 206 265″/rad gives 0.077″ = 77 mas.
2. Nyquist sampling needs pixels of $\lambda/(2D)$ = 32 mas — almost exactly the 31 mas pixels of JWST's NIRCam short-wavelength channel.
3. Hubble at 0.5 µm: θ = 1.22 × 0.5 × 10^{-6} / 2.4 = 52 mas. JWST at 2 µm is slightly coarser than Hubble in the visible, but at 0.7 µm it reaches 27 mas.
4. Ground seeing of 0.8″ is ten times worse. At the Moon's distance (384 400 km), 77 mas corresponds to 144 m; JWST could separate two car headlights 1.5 m apart at about 4 000 km.

38.2.2 Telescope Designs

Almost all space telescopes are reflectors: mirrors work from the ultraviolet to the far infrared, have no chromatic aberration and can be supported from behind. The main designs (illustrated in Chapter 12) are the **Cassegrain**, with a concave primary and a convex secondary that folds the light back through a hole in the primary. Hubble is an f/24 **Ritchey–Chrétien** Cassegrain, whose two hyperbolic mirrors remove coma over a wide field. The other main design is the **three-mirror anastigmat (TMA)**, whose third curved mirror corrects astigmatism and field curvature to give a wide, flat field. JWST, Euclid (a compact Korsch variant) and Roman are TMAs; Roman's 0.28 square-degree field is about 200 times that of Hubble's infrared camera. **Off-axis** designs tilt the mirrors so that the secondary and its struts do not block the beam. This gives a clean Airy pattern without diffraction spikes. That is why the planned Habitable Worlds Observatory (HWO) is studying an off-axis aperture for coronagraphy.

38.2.3 Mirrors: Materials and Lightweighting

A mirror must keep its shape to a small fraction of the wavelength — typically 10–30 nm rms for a visible or near-infrared telescope. It must do so while being launched at several g, cooled or heated by tens of kelvin and held without distortion in weightlessness. The material is chosen for low thermal expansion, high stiffness per unit mass and good thermal conductivity (Table 38.2).

TABLE 38.2 Typical room-temperature properties of space mirror materials.

Material	Density (g/cm³)	Stiffness (GPa)	Expansion at 20 °C (ppm/K)	Conductivity (W/m·K)	Used on
ULE glass	2.2	68	≈ 0 ± 0.03	1.3	Hubble, Roman
Zerodur glass-ceramic	2.5	90	≈ 0 ± 0.02	1.5	Chandra
Beryllium	1.85	300	11 (≈ 0 below 100 K)	216	JWST, Spitzer
Silicon carbide	3.2	420	2.2	180	Herschel, Gaia, Euclid
Aluminium alloy	2.7	69	23	167	Small and cryogenic telescopes

Glass mirrors are **lightweighted** by fusing thin face sheets to a honeycomb core. Hubble's 2.4 m primary (Figure 38.4) weighs about 828 kg instead of roughly 3 600 kg for a solid blank; Roman's mirror of the same size weighs only 186 kg. **Beryllium** is as stiff as steel but lighter than aluminium, and its expansion almost vanishes below 100 K. That is why JWST and Spitzer use it. Each JWST segment was measured at cryogenic temperature and re-polished warm with the inverse of its cold distortion. **Silicon carbide** combines stiffness and conductivity: Herschel's 3.5 m primary was brazed from 12 SiC petals and machined from 720 kg down to 240 kg.

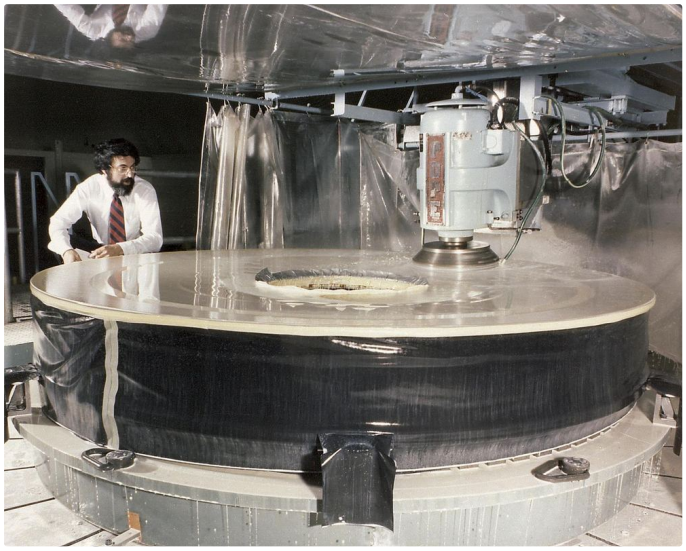


FIGURE 38.4 Hubble's 2.4 m primary mirror being ground at Perkin-Elmer in Danbury, Connecticut, around 1979. Its ultra-low-expansion glass face sheets are fused to a honeycomb core to cut its mass to about 828 kg. Image: NASA/MSFC.

Hubble also taught the most expensive lesson in optical testing. Its primary was polished very precisely — to the wrong shape. A lens in the **null corrector**, the instrument used to check the mirror's figure, was misplaced by about 1.3 mm, so the mirror's edge was about 2 µm too flat. The resulting **spherical aberration** spread most of a star's light into a wide halo. Two cross-checks with other instruments had hinted at the error and been dismissed. The lesson — test flight hardware end-to-end, with independent methods ("test as you fly") — has shaped every programme since.

38.2.4 Segmented and Deployable Mirrors

Fairing diameter limits a monolithic mirror to about 4–5 m today. JWST's 6.5 m primary is therefore made of 18 hexagonal beryllium segments, each about 1.32 m across flats and weighing about 20 kg. They are coated with about 100 nm of gold, which reflects some 99 % of infrared light. Two wings of three segments fold back for launch, and the secondary mirror is held in front of the primary by a deployable tripod.

Each segment sits on six actuators (a **hexapod**) that set its position and tilt, plus a seventh that changes its curvature, moving in steps of roughly ten nanometres (Figure 38.5). Commissioning in February–April 2022 followed a **wavefront sensing and control** sequence. First, each segment formed its own image of a bright star, and the 18 images were tilted one by one until they overlapped. Then **coarse phasing** with a dispersed Hartmann sensor in NIRCam measured and

removed height (piston) differences between segments. Finally, in **fine phasing**, NIRCam took deliberately defocused images through weak lenses, and **phase-retrieval** software computed the wavefront error across the whole pupil for the actuators to correct.

The result beat the requirement. The wavefront error at NIRCam is typically 60–80 nm rms, against a prediction of under 100 nm. The segments drift by less than 10 nm between checks every two days, with small corrections every month or two. One surprise was a micrometeoroid that hit segment C3 in May 2022, larger than pre-launch models predicted. It raised that segment's error from about 56 to more than 250 nm rms, partly corrected to 178 nm. This added about 9 nm to the telescope total. JWST now limits the time it spends looking along its direction of orbital motion, where impacts are more energetic.



FIGURE 38.5 A JWST engineering-model mirror segment with its gold coating. The actuators that position the segment and adjust its curvature are mounted behind it. Image: NASA/GSFC.

38.2.5 Grazing Incidence, Coded Masks and Pair Telescopes

X-rays penetrate a normal mirror; they reflect only at **grazing incidence**, at angles below about 1°, and the critical angle shrinks as photon energy rises. A **Wolter type I** telescope therefore uses a ring-shaped paraboloid followed by a hyperboloid, and nests several pairs to increase the collecting area (Figure 38.6). Chandra has four nested pairs of iridium-coated Zerodur shells, the largest 1.23 m across, with a 10 m focal length (Figure 38.7). Its 1 484 kg mirror assembly resolves better than 0.5″, still the sharpest of any X-ray telescope. XMM-Newton trades sharpness for area with three modules of 58 thin gold-coated nickel shells each. NuSTAR reaches 79 keV with 133 shells per optic coated with **multilayers** — alternating heavy and light layers that reflect like a crystal. The optics sit at the end of a 10 m deployable mast.

At higher energies, imaging uses shadows. Swift's Burst Alert Telescope places a **coded mask** of about 52 000 lead tiles one metre above 32 768 cadmium zinc telluride (CZT) crystals. Each source casts a shifted shadow of the mask. Software decodes the overlapping shadows to locate gamma-ray bursts within seconds. Above about 20 MeV, where a gamma ray mostly turns into matter, each photon is tracked individually. In a **pair-conversion telescope** such as Fermi's Large Area Telescope, a photon turns into an electron–positron pair in a thin tungsten foil. Silicon strip detectors track the pair, and a caesium iodide calorimeter measures its energy.

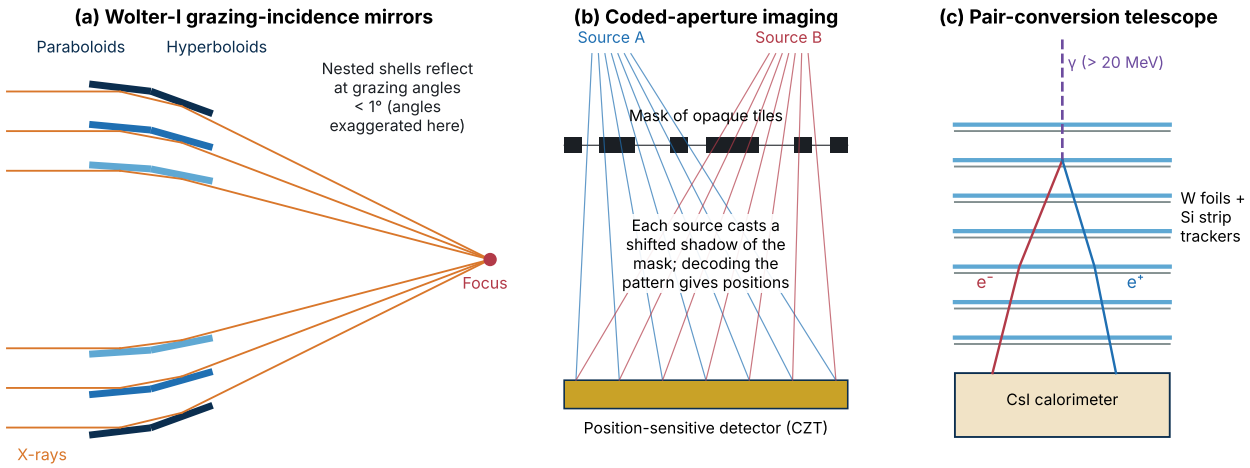


FIGURE 38.6 Optics for high-energy astronomy: (a) nested Wolter-I grazing-incidence mirrors focus X-rays after two reflections; (b) a coded mask casts a shadow pattern that encodes the direction of each source; (c) a pair-conversion telescope tracks the electron and positron created by a gamma ray.

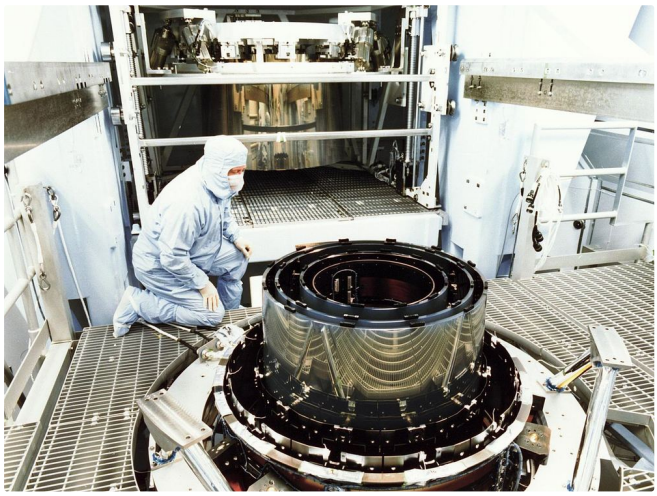


FIGURE 38.7 The nested mirror shells of Chandra's High Resolution Mirror Assembly being assembled at Eastman Kodak in Rochester, New York. Image: NASA/MSFC.

38.2.6 Blocking the Star: Coronagraphs and Starshades

An Earth-like planet is extremely faint next to its star. With a geometric albedo of 0.2 its brightness relative to the star is $A_g(R_p/a)^2 \approx 3.6 \times 10^{-10}$ at full phase and about 1.2×10^{-10} when half-lit. At 10 parsecs it lies only 0.1″ from the star — about six times λ/D for a 6 m telescope in visible light. Seeing it requires suppressing starlight ten-billion-fold close to the diffraction limit.

A **coronagraph** blocks the star's image with a small focal-plane mask, then removes the diffracted light with a **Lyot stop** in a re-imaged pupil. JWST's coronagraphs reach contrasts of order 10^{-5}, enough for young, self-luminous giant planets in the infrared. Roman's Coronagraph Instrument is the first in space to correct wavefront errors actively with two **deformable mirrors**, and aims to show contrasts of 10^{-7} or better. Reaching 10^{-10} needs a telescope roughly ten times more stable than Roman, with a wavefront steady to picometres over hours, and a coronagraph about a hundred times deeper. This is the central technology challenge for HWO, whose project office aims to mature it before a mission concept review at the end of the decade. However, NASA's FY2027 budget request (not enacted by October 2026) proposed cutting HWO's funding from US$150 million to US$5 million.

The alternative is a **starshade**: a flower-shaped screen tens of metres across, flying tens of thousands of kilometres in front of the telescope, whose petals diffract starlight away while the planet's light passes by (Figure 38.8). Laboratory masks have reached 10^{-10} contrast, but a starshade must travel for days to weeks between target stars. NASA closed out its starshade technology activity in April 2025, and HWO's baseline is a coronagraph.

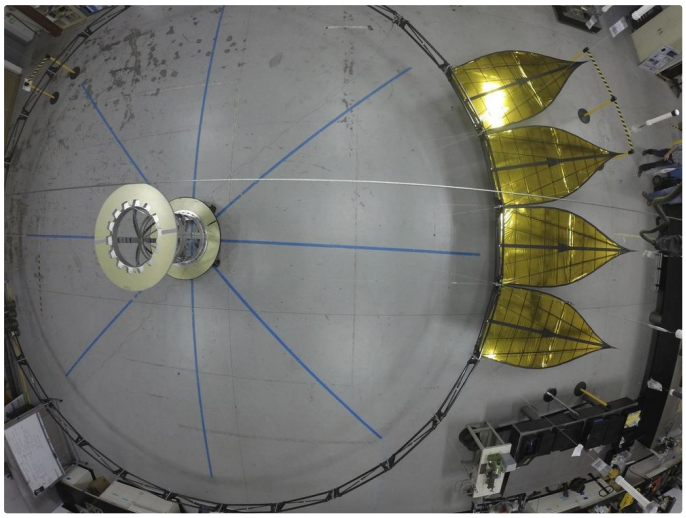


FIGURE 38.8 A half-scale starshade with four petals deployed at NASA's Jet Propulsion Laboratory in 2014. The precisely shaped petal edges diffract starlight away from the telescope. Image: NASA/JPL (NASA/JPL-Caltech/Princeton).

38.3 Detectors

38.3.1 From Photons to Electrons

A detector converts photons into electrical charge and measures it. Its key properties are **quantum efficiency (QE)** — the fraction of photons that produce a detected electron — **read noise**, **dark current** (charge generated thermally in darkness), format and radiation tolerance (Table 38.3).

Charge-coupled devices (CCDs) store photoelectrons in potential wells and shift them pixel by pixel to one low-noise amplifier, reaching QE above 90 % and read noise of 2–4 electrons. Their weakness in space is **radiation damage**: proton-induced traps capture charge as it is shifted, smearing images. Hubble's CCDs lose transfer efficiency every year, and its pipeline corrects each pixel with a model of the traps. **CMOS** sensors put an amplifier in every pixel, read out faster, need less power and tolerate radiation better.

TABLE 38.3 Detectors used in space astronomy.

Detector	Range	Temperature	Typical format	Examples
Silicon CCD	0.2–1.0 µm	150–190 K	4k × 4k	Hubble, Gaia (106 CCDs), Euclid VIS (36)
Silicon CMOS	0.3–1.0 µm	200–300 K	Up to 4k × 4k	Star trackers, guide cameras, small telescopes
HgCdTe (H2RG, H4RG)	0.6–5.3 µm	35–100 K	2k × 2k, 4k × 4k	JWST, Euclid NISP, Roman WFI

Detector	Range	Temperature	Typical format	Examples
Si:As impurity band	5–28 µm	< 7 K	1k × 1k	JWST MIRI, Spitzer, WISE
Bolometers	60 µm–3 mm	0.1–0.3 K	10^2–10^3 pixels	Herschel SPIRE, Planck HFI
X-ray CCD	0.2–12 keV	≈ 150 K	1k × 1k	Chandra ACIS, XMM-Newton EPIC
Microcalorimeter	0.3–12 keV	0.05 K	6 × 6	XRISM Resolve
CZT	3–150 keV	≈ 290 K	10^4 crystals	Swift BAT, NuSTAR

Infrared photons have too little energy to free electrons in silicon. **Mercury cadmium telluride (HgCdTe)**, whose band gap is set by its composition, is bonded pixel by pixel with indium bumps to a silicon readout circuit. Teledyne's **H2RG** (2048 × 2048 pixels of 18 µm) serves JWST's near-infrared instruments and Euclid; Roman uses 18 **H4RG-10** arrays (4096 × 4096 pixels of 10 µm), about 300 million pixels (Figure 38.9). These arrays are read non-destructively many times "up the ramp" during an exposure, which averages down read noise and lets software remove cosmic-ray hits as sudden jumps. Beyond 5 µm, MIRI uses arsenic-doped silicon **blocked-impurity-band** detectors.

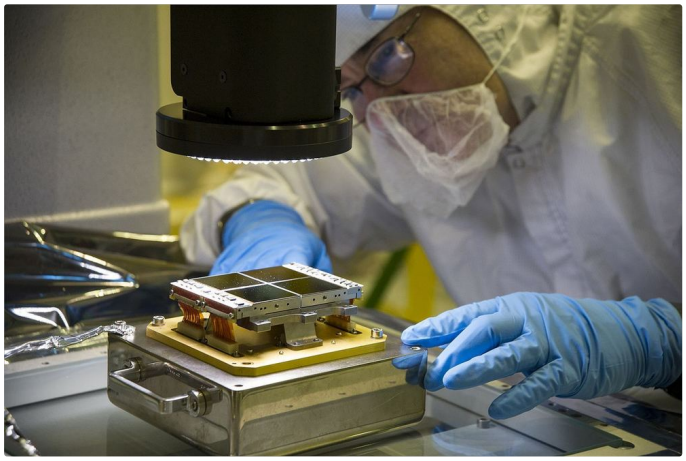


FIGURE 38.9 A replacement focal-plane assembly for JWST's Near Infrared Camera being aligned under a three-dimensional microscope at NASA Goddard. Each assembly holds HgCdTe detector arrays on a precisely flat mount. Image: NASA/GSFC.

38.3.2 Bolometers and Microcalorimeters

At far-infrared and millimetre wavelengths, detectors measure the **heat** deposited by radiation. A **bolometer** is a tiny absorber on a weak thermal link to a cold bath, with a sensitive thermometer — a doped semiconductor or a **transition-edge sensor** held on the edge of superconductivity. Planck cooled its bolometers to 0.1 K and mapped the cosmic microwave background to millionths of a kelvin. **Kinetic inductance detectors**, superconducting resonators read by the thousand on one wire, are being developed for future far-infrared missions.

An X-ray **microcalorimeter** measures the heat of single photons. XRISM's Resolve, launched in 2023, has a 6 × 6 array at 50 mK, cooled by adiabatic demagnetisation refrigerators. The tiny temperature rise from each photon gives its energy to about 4.5 eV at 6 keV — some 30 times finer than an X-ray CCD. This reveals the motions of hot gas in galaxy clusters. Its protective gate valve

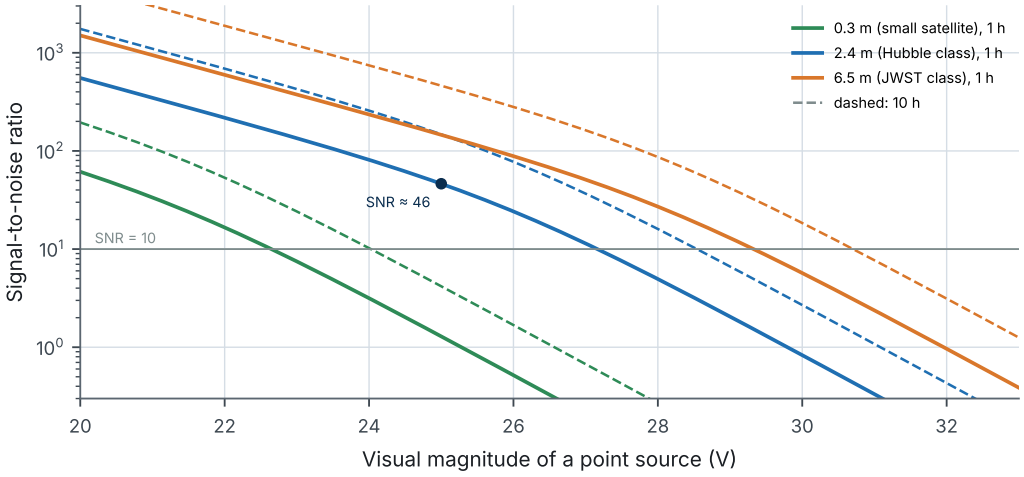
has failed to open, limiting it to energies above about 1.7 keV. Hard X-rays and gamma rays are absorbed in thick, dense semiconductors such as **CZT**, which work at room temperature, or in scintillating crystals.

38.3.3 Noise and the Photon Budget

Whether a source is detectable depends on how many photons arrive compared with the noise. The **signal-to-noise ratio (SNR)** of a point source measured in an aperture of n_{pix} pixels is

$$\mathrm{SNR} = \frac{S}{\sqrt{S + n_{pix}\left(B + D\,t + n_{read}\,\sigma_{read}^{2}\right)}}$$

where S is the number of source electrons, B the sky electrons per pixel, D the dark current, t the exposure time and σ_{read} the read noise for each of n_{read} readouts. Faint sources are **background-limited**: SNR grows only as $\sqrt{t}$, so a tenfold longer exposure gains 1.25 magnitudes. A larger telescope helps twice — it collects more light, and its smaller image includes less sky. In one hour, a 2.4 m telescope reaches SNR = 10 at V ≈ 27.2 and a 6.5 m telescope at V ≈ 29.3 (Figure 38.10).



Model: V band (550 nm, 88 nm wide), 30 % throughput, 33 % obstruction, diffraction-limited aperture r = 1.22 λ/D holding 80 % of the light, zodiacal sky 22.7 mag/arcsec², Nyquist pixels, dark 0.005 e⁻/s, read noise 3 e⁻ per 1 000 s sub-exposure.

FIGURE 38.10 Signal-to-noise ratio versus magnitude for 0.3 m, 2.4 m and 6.5 m diffraction-limited space telescopes observing in the visible for 1 hour (solid) and 10 hours (dashed). The dot marks the worked example.

EXAMPLE: HOW MANY PHOTONS FROM A 25TH-MAGNITUDE STAR?

A V = 25 star is observed for one hour with a 2.4 m telescope (33 % central obstruction) through a V filter (550 nm, 88 nm wide) with 30 % overall throughput.

1. A V = 0 star delivers 3.63×10^{-11} W m^{-2} nm^{-1}. Each 550 nm photon carries $hc/\lambda = 3.61 \times 10^{-19}$ J, so the flux is 1.0×10^{8} photons s^{-1} m^{-2} nm^{-1}.
2. V = 25 is 10^{-10} times fainter: 0.88 photons s^{-1} m^{-2} across the 88 nm band.
3. Collecting area $(\pi/4) \times 2.4^{2} \times (1 - 0.33^{2}) = 4.03$ m^{2} gives 3.6 photons/s; with 30 % throughput, 1.07 electrons/s, or 3 850 per hour, of which 80 % (3 080) fall in the measuring aperture.
4. Noise: zodiacal sky (22.7 mag/arcsec^{2}) adds 335 electrons, dark current 337 and read noise 673 (19 pixels, four readouts of 3 electrons). Total noise = √(3 080 + 335 + 337 + 673) = 66.5 electrons.
5. SNR = 3 080 / 66.5 ≈ **46**. A star some 40 million times fainter than the faintest visible to the naked eye (V ≈ 6) yields about one detected photon per second on a Hubble-sized telescope.

38.4 Keeping Cold and Clean

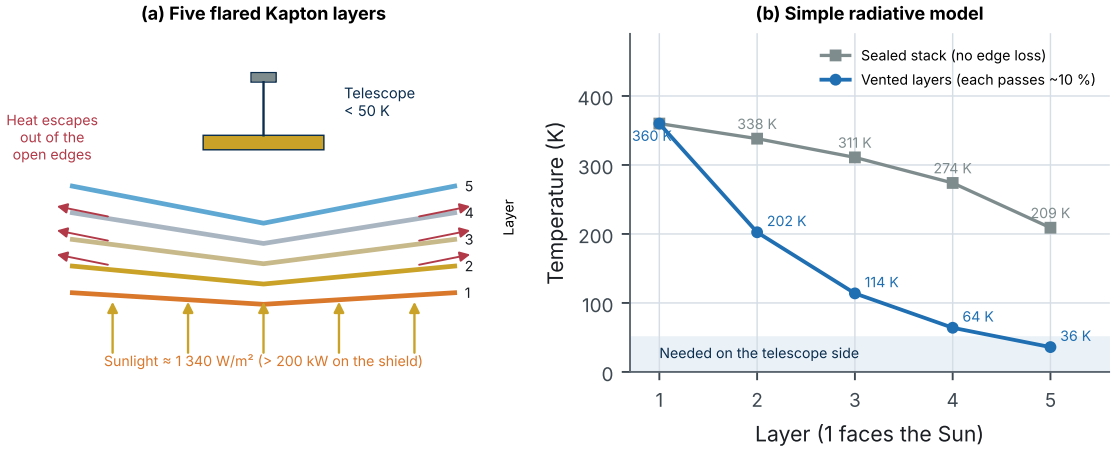
38.4.1 Passive Cooling and Sunshields

The simplest way to cool a telescope is to let it radiate to deep space while shading it from the Sun, Earth and Moon. Radiation scales with the fourth power of temperature, so a 40 K surface can reject only 0.15 W per square metre even with perfect emissivity. Every milliwatt leaking from the warm spacecraft must therefore be intercepted.

JWST's **sunshield** is the extreme example (Figure 38.11). Five membranes of Kapton E, the first 50 µm thick and the rest 25 µm, span about 21 m by 14 m. They are coated with about 100 nm of aluminium and, on the two Sun-facing layers, doped silicon. More than 200 kW of sunlight falls on the first layer, which reaches up to about 383 K in NASA's thermal models. Only about 23 mW reaches the telescope side, where the last layer is near 36–40 K. The layers never touch, and they are **flared** — closer at the centre and wider apart at the edges — so that heat radiated into each gap escapes sideways to space (Figure 38.12).



FIGURE 38.11 JWST's full-size sunshield test unit stacked and unfurled in a cleanroom. The five membranes, each thinner than a human hair, are separated by gaps that let heat escape out of the edges. Image: NASA/GSFC.



Model: layer 1 at 360 K; sealed: four gaps with ε = 0.05 faces, layer 5 radiating to space; vented: flux falls ×10 per layer (reported for JWST). NASA quotes ≈383 K maximum and ≈36 K.

FIGURE 38.12 (a) The five flared sunshield layers between the Sun and the telescope. (b) Layer temperatures in two simple radiative models: a sealed stack, in which heat can only pass through the layers, and vented layers, each of which passes on about a tenth of the heat it receives.

EXAMPLE: WHY THE SUNSHIELD GAPS MUST BE OPEN

At L2, sunlight delivers S = 1 361 / 1.01² = 1 334 W/m², so the 200 kW on JWST's first layer implies about 150 m² facing the Sun.

1. A plate absorbing all this sunlight and radiating from one face reaches $(S/\sigma)^{1/4}$ = 392 K — close to the first layer's 383 K maximum.
2. **Sealed stack.** If the five layers formed a closed sandwich with low-emissivity faces (ε = 0.05), each of the four gaps would have a radiative resistance of $(2/\varepsilon - 1)$ = 39. With the first layer at 360 K and the last at 40 K, the heat flow is $q = \sigma(360^4 - 40^4)/(4\times39)$ = 6.1 W/m², about 900 W in all.
3. To radiate 6.1 W/m², the cold layer would have to warm to at least 102 K even with perfect emissivity — far too warm for a mid-infrared telescope.
4. **Vented layers.** The real leak, 23 mW over 150 m², is 0.15 mW/m² — about 40 000 times less. Suppose each vented layer passes on a tenth of the heat it receives. Then four steps reduce the radiated flux by 10^{-4} and the temperature by $(10^{-4})^{1/4}$ = 0.1: from 360 K to 36 K, as Figure 38.12 shows.

38.4.2 Cryocoolers and Cryogens

Passive cooling stops at about 30–40 K for a large telescope near Earth. Early infrared missions carried **cryostats** of liquid helium that slowly boiled away. Spitzer's lasted from 2003 to 2009, after which it continued a "warm" mission at about 30 K. Herschel's 2 300 litres of superfluid helium ran out in April 2013, ending its science. Spitzer pioneered the **warm launch**: only the instruments sat inside the cryostat, and the telescope cooled in space by radiating from its black outer shell.

Mechanical **cryocoolers** do not run out. JWST's MIRI detectors must stay below 7 K. Its cooler (Figure 38.13) uses a three-stage **pulse-tube** precooler, driven by a compressor in the warm spacecraft bus, to chill helium gas to about 14 K. The helium then flows about 10 m to a **Joule–Thomson** stage beside MIRI. There, expansion through an orifice less than 1 mm wide cools it to about 6 K. The only moving parts are horizontally opposed pistons, balanced so as not to shake the telescope. For 0.1 K and below, Planck used a dilution refrigerator and XRISM uses magnetic refrigeration.

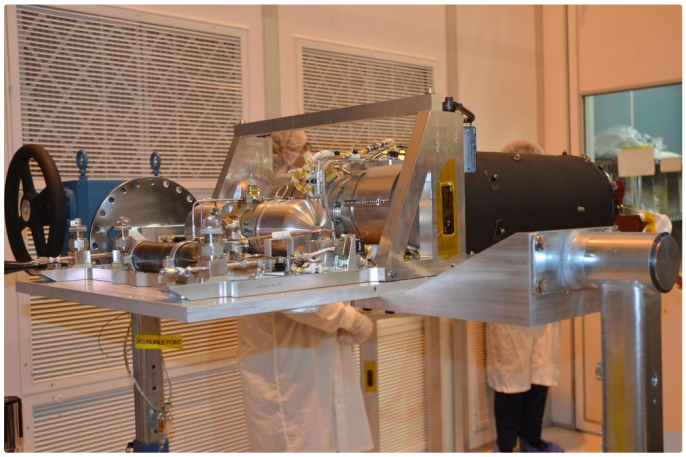


FIGURE 38.13 The flight cryocooler for JWST's Mid-Infrared Instrument after testing in 2016, before it was shipped to Northrop Grumman for installation on the spacecraft. It cools MIRI's detectors to below 7 K, about 30 K colder than the rest of the telescope. Image: NASA/JPL (NASA/JPL-Caltech).

38.4.3 Stray Light and Contamination

Stray light is sunlight or earthshine scattered off baffles and struts, light from bright stars outside the field, or glow from warm structures. It is controlled with baffles, black coatings and field stops. These are designed by tracing millions of rays through a model of the telescope (Chapter 12). Mirror microroughness of a few nanometres scatters light into a halo that hides faint objects near bright ones.

Contamination is the other enemy. Molecules **outgassing** from adhesives, cables and insulation condense on cold optics; a film a few nanometres thick absorbs far-ultraviolet light. Space materials are screened for outgassing (typically below 1 % total mass loss and 0.1 % condensable material), baked in vacuum and assembled in cleanrooms under dry-nitrogen purge. Water is the hardest contaminant to eliminate. After **Euclid**'s launch in July 2023 its throughput fell steadily as water released by insulation condensed as ice on its cold mirrors. In March 2024 ESA warmed a mirror behind the main optics from about 126 K to 160 K for under two hours, and repeated the procedure in June. The visible camera regained about 15 % more light, and throughput has been stable since (Figure 38.14).



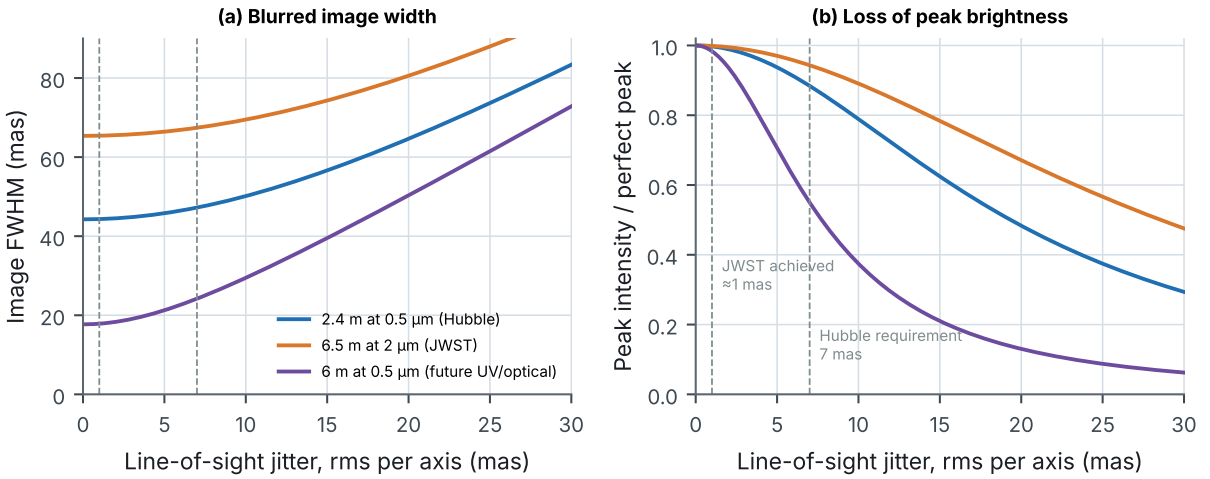
FIGURE 38.14 ESA's Euclid being secured to the adapter of its Falcon 9 launcher in June 2023. The white cylinder houses the 1.2 m silicon-carbide telescope; the black solar panels also shade it from the Sun. Image: NASA/JPL (ESA).

38.5 Pointing and Stability

38.5.1 Sensing and Control

Coarse pointing uses sun sensors, star trackers and gyroscopes, as on any spacecraft (Chapter 6). The final step is a **fine guidance sensor (FGS)** that measures a guide star's position in the telescope's own focal plane, so that flexing between a separate star tracker and the instrument does not matter. Hubble's three interferometric FGS and four **reaction wheels** hold a target to about 7 mas rms over 24 hours. JWST's FGS reads guide stars 16 times a second and drives a **fine steering mirror**; in orbit its jitter is about 1 mas per axis, several times better than required.

Jitter blurs the image and lowers its peak brightness (Figure 38.15). A useful rule is that it should stay below about a tenth of λ/D: 4–5 mas for Hubble at 0.5 µm, and about 2 mas for a future 6 m ultraviolet–optical telescope. Coronagraphy is stricter still, because drift moves starlight off the mask; HWO studies discuss sub-milliarcsecond pointing together with picometre wavefront stability.



Gaussian jitter convolved with the diffraction core: $\mathrm{FWHM}^2 = (1.03\,\lambda/D)^2 + (2.355\,\sigma)^2$; peak ratio $\approx 1/[1 + (\pi^2/2)(\sigma D/\lambda)^2]$.

FIGURE 38.15 Effect of line-of-sight jitter on (a) image width and (b) peak brightness for three telescopes. Hubble's 7 mas requirement and JWST's measured jitter of about 1 mas are marked.

Most jitter comes from inside the spacecraft. **Reaction wheels** spinning at thousands of revolutions per minute have tiny imbalances that shake the structure; cryocooler compressors, **thermal snaps** and flexing solar arrays add more. Hubble's first solar arrays vibrated every time the telescope crossed from shadow into sunlight, until stiffer arrays were fitted in 1993 and 2002. Engineers balance wheels and choose speeds that avoid structural resonances. They mount wheels or whole telescopes on soft **isolators** (JWST has a passive isolator between its bus and telescope). They also stiffen and damp the structure, and close fast optical loops with steering mirrors.

38.5.2 Hubble's Gyroscopes

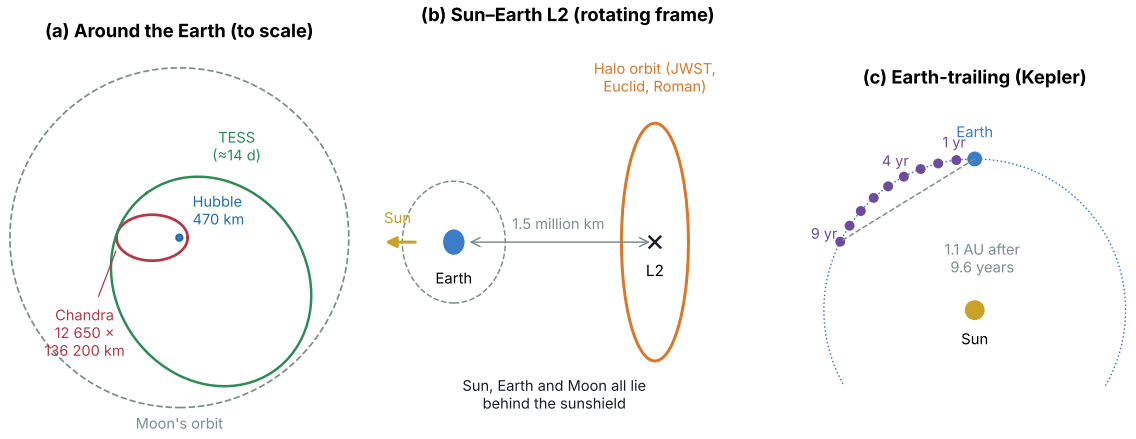
Hubble's gyroscopes sense rotation through a spinning wheel floating in fluid, connected by thin wires called **flex leads** that corrode; by 2023, 8 of the 22 gyros flown had failed. Hubble carries six and normally used three. Servicing Mission 4 replaced all six in 2009 — the last time this was possible. After a gyro repeatedly gave faulty readings in 2024, NASA switched Hubble to **one-gyro mode** in June 2024, keeping its other working gyro in reserve. Magnetometers, sun sensors and star trackers fill the gaps during slews. NASA expected about 12 % lower efficiency (74 instead of 85 orbits of observing a week) and slower target acquisition. It also expected less sky available at any time and no tracking of targets closer than Mars. Image quality is unchanged. NASA estimated a better than 70 % chance of at least one gyro working through 2035; by late 2025 the observing success rate was back above 90 %, with no safe-mode events since the switch.

38.5.3 Scanning Telescopes

Not every telescope stares. **Gaia** spun once every six hours, sweeping two fields of view across the sky. As it spun, its 106 CCDs moved charge in step with the star images — **time-delay integration**, as in pushbroom Earth imagers (Chapter 14). Over 10.5 years it measured positions of nearly two billion stars to tens of microarcseconds. Planck spun once a minute to map the microwave background, and WISE and SPHEREx scanned the whole sky. Scanning simplifies attitude control but demands precise knowledge of the spin axis and a readout synchronised to the motion.

38.6 Where to Put a Telescope

The orbit sets a telescope's thermal environment, observing efficiency, data rate and whether it can be serviced (Figure 38.16, Table 38.4).



(a) to scale; (b) schematic; (c) circular-orbit approximation of Kepler's 372.5-day heliocentric orbit (lag ≈ 7° per year), shown in a frame rotating with the Earth.

FIGURE 38.16 Telescope orbits: (a) low Earth orbit (Hubble), Chandra's highly elliptical orbit and TESS's lunar-resonant orbit, to scale; (b) a halo orbit around Sun–Earth L2 (schematic); (c) the slow drift of an Earth-trailing telescope such as Kepler, in a frame rotating with the Earth.

Low Earth orbit is cheap to reach and was the only orbit the Space Shuttle could service, so Hubble flies at about 470 km. But Earth hides most targets for part of every 95-minute orbit. The telescope crosses the **South Atlantic Anomaly**, where trapped protons swamp the detectors, and it passes from sunlight to shadow about 15 times a day. Heaters hold Hubble's mirrors at about 21 °C. **Highly elliptical orbits** give long, uninterrupted views: Chandra's 64-hour orbit, now about 12 650 × 136 200 km, keeps it above the radiation belts for most of each orbit. TESS uses a stable 2:1 resonance with the Moon.

Sun–Earth L2, 1.5 million km beyond Earth on the anti-Sun side, has become the standard home of large infrared and survey telescopes: Herschel, Planck, Gaia, JWST, Euclid and, from December 2026, Roman. The Sun, Earth and Moon always lie in roughly the same direction, so one sunshield blocks all three, and the thermal environment is almost constant. L2 is unstable, so telescopes fly **halo orbits** around it with small station-keeping burns every few weeks; JWST's accurate launch left propellant for more than 20 years. **Earth-trailing** heliocentric orbits, used by Spitzer and Kepler, need no station-keeping, but the telescope drifts away — Kepler fell behind by about 7° a year — so its radio link weakens with time.

TABLE 38.4 Trade-offs between orbits for space telescopes.

Orbit	Examples	Thermal	Observing efficiency	Downlink	Servicing
LEO	Hubble, Swift, Fermi	Cycles every 95 min	≈ 50 %	Relay satellites, ground passes	Possible
Highly elliptical	Chandra, XMM-Newton, TESS	Long stable arcs	High; day-long exposures	Ground stations	No
Sun–Earth L2	JWST, Euclid, Gaia, Roman	Nearly constant	Very high	Deep-space antennas	Not designed for it
Earth-trailing	Spitzer, Kepler	Very stable, cold	Very high	Weakens with distance	No

38.7 Building, Testing and Servicing

38.7.1 Test as You Fly

A telescope that cannot be repaired is tested on the ground at the temperatures and in the configuration it will fly. Complete observatories are shaken on vibration tables, exposed to the acoustic roar of launch and tested in **thermal-vacuum** chambers. JWST's telescope and instruments spent about 100 days in 2017 inside Chamber A at NASA's Johnson Space Center — a 36 m tall chamber built for Apollo. They were cooled by liquid-nitrogen and helium shrouds while optical tests confirmed that the segments could be aligned at operating temperature (Figure 38.17). Gravity sag, which cannot be removed on the ground, is predicted with finite-element models and checked by testing mirrors face-up and face-down. JWST's deployments were rehearsed many times; after launch on 25 December 2021, its roughly 50 major deployments took two weeks and all worked.

38.7.2 Hubble Servicing Missions

Hubble was designed to be serviced, with handholds, standard bolts and **orbital replacement units** — instruments, batteries, gyroscopes, computers — that astronauts could exchange during spacewalks from the Space Shuttle (Chapter 33). Five missions between 1993 and 2009 corrected its optics, replaced all of its original instruments and extended its life by decades (Table 38.5).

TABLE 38.5 The five Hubble servicing missions.

Mission	Date	Shuttle	Main work
SM1	Dec 1993	Endeavour (STS-61)	COSTAR corrective optics; WFPC2; new solar arrays and gyros
SM2	Feb 1997	Discovery (STS-82)	STIS spectrograph; NICMOS infrared camera; solid-state recorder
SM3A	Dec 1999	Discovery (STS-103)	All six gyros after Hubble entered safe mode; new computer
SM3B	Mar 2002	Columbia (STS-109)	Advanced Camera for Surveys; rigid solar arrays; NICMOS cryocooler
SM4	May 2009	Atlantis (STS-125)	WFC3 and COS; repair of ACS and STIS; six gyros; batteries

On the first mission astronauts installed COSTAR — small deployable mirrors that refocused light for three instruments — and replaced the Wide Field/Planetary Camera with WFPC2, whose internal optics compensated for the mirror error (Figure 38.18). Without shuttle visits, Hubble's orbit is slowly decaying, from more than 600 km at its launch in 1990 to about 470 km in 2026. NASA has studied but not ordered a commercial reboost; without one, STScI's median forecast for re-entry is late 2033.

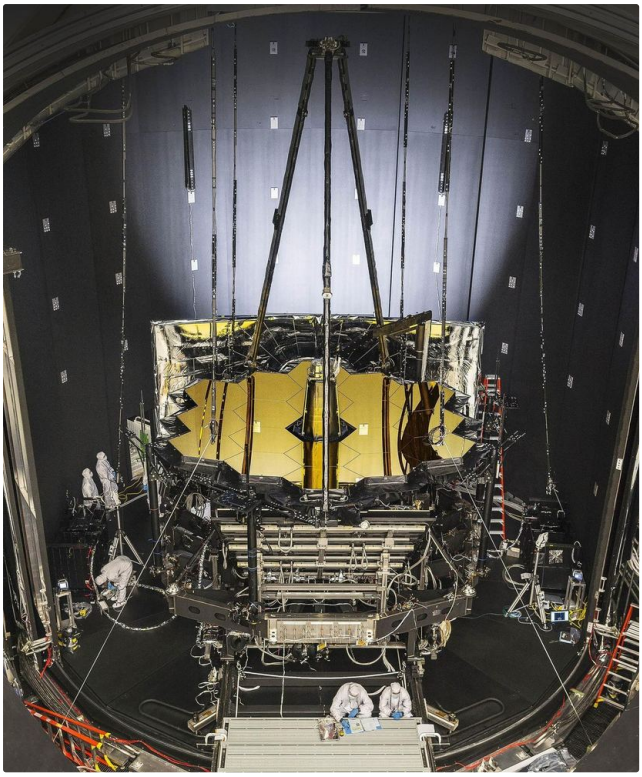


FIGURE 38.17 JWST's telescope and science instruments inside Chamber A at NASA's Johnson Space Center in 2017, before cryogenic vacuum testing that simulated the cold of space. Image: NASA/GSFC.



FIGURE 38.18 Astronaut Jeffrey Hoffman holds the original Wide Field/Planetary Camera in Endeavour's payload bay during the first Hubble servicing mission in December 1993, after replacing it with WFPC2. Image: NASA/MSFC.

38.7.3 Assembling the Next Generation in Space

JWST shows how far folding can go: a 6.5 m aperture inside a 5 m fairing. Apertures of 10–20 m may need to be assembled in orbit from separately launched segments, by robots or astronauts, and aligned with the same wavefront sensing used on JWST. HWO studies include standard interfaces and an orbit a future servicing vehicle could reach, so that instruments could be upgraded as Hubble's were (Chapter 40).

38.8 From Photons to Archives

38.8.1 Downlink and Storage

Data volume grows with detector size, and the downlink must keep pace (Table 38.6). Hubble sends about 18 GB of science data a week through NASA's Tracking and Data Relay Satellites. JWST stores data on a 68 GB solid-state recorder. It sends up to about 57 GB a day to the Deep Space Network's 34 m antennas on a 25.9 GHz Ka-band link. At up to 28 Mbit/s, this takes about

4.5 hours of transmission. Euclid returns about 100 GB of compressed data a day in a four-hour pass. Roman will send about 1.4 TB a day at up to 500 Mbit/s to stations in New Mexico, Australia and Japan — more than Hubble returns in a year.

TABLE 38.6 Five major space telescopes and their data rates.

Telescope	Launched	Aperture	Band	Orbit	Data per day
Hubble	1990	2.4 m	0.115–1.7 µm	LEO, 470 km	≈ 2.6 GB
Chandra	1999	1.2 m (outer shell)	0.1–10 keV	12 650 × 136 200 km	Small (photon lists)
JWST	2021	6.5 m	0.6–28.5 µm	L2 halo	≈ 57 GB (28 Mbit/s)
Euclid	2023	1.2 m	0.55–2.0 µm	L2 halo	≈ 100 GB (≈ 75 Mbit/s)
Roman	2026	2.4 m	0.48–2.3 µm	L2 halo	≈ 1.4 TB (≤ 500 Mbit/s)

38.8.2 Pipelines and Archives

Raw data are useless until calibrated. Each observatory runs a **pipeline**. For JWST, stage 1 fits the slope of each pixel's ramp and removes cosmic rays, and stage 2 applies flat fields, flux calibration and sky coordinates. Stage 3 combines dithered exposures into mosaics, spectra and catalogues. Data are reprocessed as calibrations improve. The **Mikulski Archive for Space Telescopes (MAST)** at the Space Telescope Science Institute holds data from about 20 NASA missions, including Hubble, JWST and TESS, and soon Roman. It passed 3 petabytes in 2017 and has grown rapidly since. Copies are kept in commercial clouds so that researchers can analyse data where they are stored. Most observations become public within a year, and archival research now produces a large share of published papers — the archive, not the telescope, is the observatory's lasting product.

AI IN SPACE TELESCOPE OPERATIONS

Machine learning appears throughout a space telescope's life (Chapter 51). Neural networks trained on simulated defocused star images can estimate wavefront errors directly, complementing iterative phase retrieval. Scheduling has used artificial intelligence since Hubble's SPIKE planner, built at STScI in the late 1980s around constraint-based search; modern schedulers juggle thousands of observations and their constraints. In data reduction, deep-learning models flag cosmic rays and detector artefacts, classify galaxy shapes and find rare objects among billions. Euclid's first quick data release combined neural networks with citizen scientists to search for strong gravitational lenses. The challenge is trust: every automated decision must be validated against physics and be reproducible from the archive (Chapter 53).

KEY TAKEAWAYS

- The atmosphere absorbs gamma rays, X-rays, ultraviolet and most infrared light, blurs images and glows; space offers the whole spectrum, diffraction-limited images and a dark, stable sky.
- Resolution is set by 1.22 λ/D and sensitivity by collecting area and background. JWST resolves 77 mas at 2 μm, and a 2.4 m telescope reaches V ≈ 27 at SNR = 10 in an hour.
- Large apertures use lightweighted glass, beryllium or silicon carbide, and segmented mirrors aligned in orbit to tens of nanometres; X-ray telescopes need nested grazing-incidence mirrors, gamma rays coded masks or pair-conversion trackers.
- Infrared telescopes must be cold: JWST's edge-vented, five-layer sunshield cuts 200 kW of sunlight to about 23 mW, and a cryocooler takes MIRI below 7 K.
- Fine guidance sensors, quiet wheels, isolation and steering mirrors hold pointing to milliarcseconds; Hubble has operated on one gyroscope since 2024.
- Sun–Earth L2 gives large telescopes a cold, stable home at the cost of servicing; Hubble's five servicing missions showed the value of designing for repair.
- Data rates have grown from gigabytes a week (Hubble) to terabytes a day (Roman); pipelines and archives matter as much as telescopes.

REVIEW QUESTIONS

1. Why must X-ray and far-infrared astronomy be done above the atmosphere? What does a space telescope gain in the visible?
2. Calculate the diffraction limit of a 1.2 m telescope at 0.8 μm and at 2 μm in milliarcseconds, and the pixel size that Nyquist-samples each.
3. Using Figure 38.2, explain why MIRI is limited by the telescope's own emission at its longest wavelengths. What would cooling the telescope to 10 K gain?
4. Compare beryllium, ULE and SiC mirrors for a warm UV and a 40 K infrared telescope.
5. Describe the steps used to align JWST's 18 segments in orbit. Why could the alignment not be completed on the ground?
6. Why do X-ray telescopes use grazing incidence and nested mirror shells?
7. A 6 m telescope observes an Earth twin at 15 pc at 0.5 μm. How many multiples of λ/D separate planet and star, and what contrast is needed?
8. Repeat the photon-budget example for a 6.5 m telescope. Why does the SNR improve by more than the ratio of diameters?
9. Use the sealed-stack calculation to explain why JWST's sunshield gaps are open.
10. Compare LEO, a highly elliptical orbit and Sun–Earth L2 for a new ultraviolet–optical telescope that should be serviceable.

CHAPTER

39 Great Observatories and Deep-Space Telescopes



Image: NASA/JPL (NASA)

Most of the light in the Universe never reaches the ground. Space telescopes recover the gamma rays, X-rays, ultraviolet and infrared that the atmosphere absorbs, see sharper than turbulent air allows and measure brightness steadily enough to find Earth-sized planets. This chapter tells the story of NASA's four Great Observatories — Hubble, Compton, Chandra and Spitzer — and of the James Webb Space Telescope, which by 2026 had seen galaxies 280 million years after the Big Bang and analysed the air of distant worlds. It explains redshift, look-back time and the several meanings of cosmic distance, reviews the survey, exoplanet, high-energy and cosmic-microwave missions, and describes the fleet operating in October 2026 and the observatories planned for the 2030s and 2040s.

IN THIS CHAPTER

39.1 Why Observe from Space
39.2 NASA's Great Observatories
39.3 The James Webb Space Telescope
39.4 Reading the Distant Universe: Redshift, Time and Distance
39.5 Surveys and Exoplanet Hunters
39.6 High-Energy and Time-Domain Astronomy
39.7 Microwaves, Radio and Gravitational Waves
39.8 The Fleet in 2026 and the Road Ahead

An astronomer on the ground looks out through a thick, warm, moving filter. Earth's atmosphere absorbs all gamma rays and X-rays, almost all ultraviolet light and most of the infrared. Water vapour glows brightly in the infrared. Turbulence smears every star into a shimmering blob about one arcsecond (1″, 1/3 600 of a degree) across. A telescope above the atmosphere escapes all three problems. It can observe at any wavelength, reach the sharpness set by diffraction, and measure a star's brightness to a few parts per million for weeks at a time.

This chapter covers the missions, capabilities and results of space observatories, from NASA's **Great Observatories** to the fleet of October 2026. How they are engineered — mirrors, detectors, cryogenics and pointing — is the subject of Chapter 38.

39.1 Why Observe from Space

39.1.1 The Atmospheric Filter

Figure 39.1 shows which parts of the electromagnetic spectrum reach the ground. Only two broad **windows** are open: the optical window from about 0.3 to 1 µm, with partial windows in the near-infrared, and the radio window from about 1 cm to 15 m. Gamma rays and X-rays are absorbed high in the atmosphere, ultraviolet by ozone and infrared by water vapour and carbon dioxide; the ionosphere reflects long radio waves. Balloons and sounding rockets (Chapters 8 and 9) sample some bands briefly; only satellites observe for years.

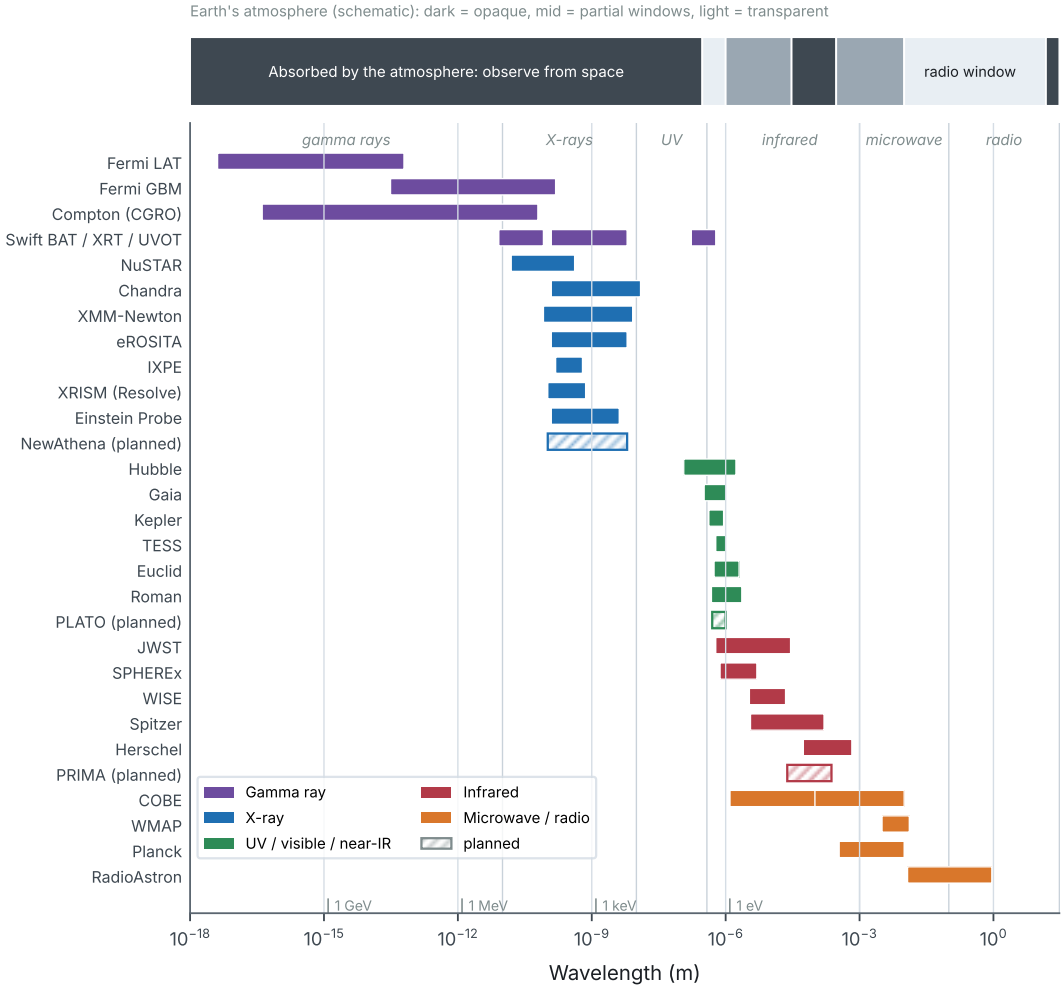


FIGURE 39.1 Wavelength coverage of the main space observatories, past, present and planned. The strip at the top shows schematically where Earth's atmosphere is opaque, partly transparent or clear: gamma-ray, X-ray, ultraviolet and far-infrared astronomy is possible only from space.

39.1.2 Sharpness, Darkness and Stability

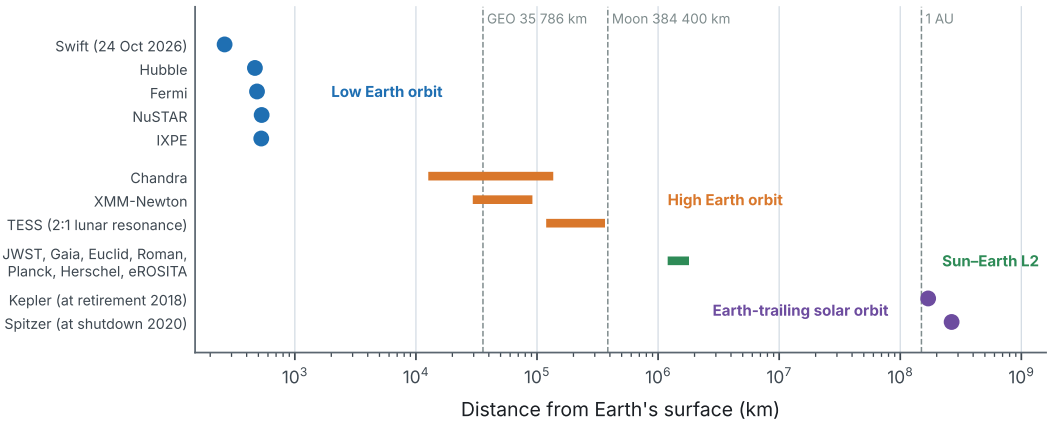
Even in the open windows, space wins three ways.

- **Resolution.** A perfect telescope of diameter D observing at wavelength λ cannot separate points closer than about $\theta = 1.22\,\lambda/D$ radians, the **diffraction limit**. For Hubble's 2.4 m mirror at 0.5 µm this is 0.05″, twenty times sharper than typical ground-based seeing. Adaptive optics now correct turbulence on large ground telescopes in the near-infrared, but only over small fields.
- **Background.** Airglow in the upper atmosphere makes the night sky glow, and in the infrared the atmosphere and the telescope itself shine like a warm oven. A telescope cooled to 40 K in deep space has an infrared background many orders of magnitude lower. This is why JWST, with a third of the collecting area of the largest ground telescopes, is far more sensitive than any of them beyond about 2 µm.
- **Stability.** Starlight twinkles (scintillation). Above the atmosphere a star's brightness can be measured continuously to tens of parts per million, the precision needed to see an Earth-sized planet cross a Sun-like star.

The price is high. A space telescope must fit inside a fairing about 5 m across, survive launch and work without adjustment by hand. It can rarely be repaired. Its cost per square metre of collecting area is hundreds of times that of a ground telescope. The two approaches are complementary: giant ground telescopes such as the 39 m Extremely Large Telescope under construction in Chile collect more light, while space telescopes provide wavelengths, sharpness and stability that the ground cannot.

39.1.3 Where Space Telescopes Fly

The orbit trades access, thermal environment and observing efficiency (Figure 39.2). **Low Earth orbit (LEO)**, used by Hubble, Fermi, NuSTAR and IXPE, is cheap to reach and shielded by Earth's magnetic field. But Earth hides most targets for part of every 95-minute orbit and warms the spacecraft. X-ray telescopes such as Chandra use **highly elliptical orbits** (HEO) reaching a third of the way to the Moon, above the radiation belts for most of the time. The cold, quiet observatories — WMAP, Planck, Herschel, Gaia, JWST, Euclid and Roman — use **halo or Lissajous orbits around the Sun–Earth L2 point**, 1.5 million km beyond Earth (Chapter 3). There the Sun, Earth and Moon always lie on one side, and one sunshield can block all three. Kepler and Spitzer used **Earth-trailing solar orbits**, drifting slowly away; when they were switched off they were 170 and 266 million km from Earth.



Bars: perigee–apogee (Earth orbits, CelesTrak elements Oct 2026) or approximate range of distance for L2 halo orbits. Dots: distance at end of mission (JPL Horizons).

FIGURE 39.2 Where astronomical observatories fly, from the low orbits of Hubble and the X-ray and gamma-ray missions to the Sun–Earth L2 point and Earth-trailing solar orbits.

39.2 NASA's Great Observatories

In the 1980s NASA planned four large observatories, each covering a different part of the spectrum. They were the **Hubble Space Telescope** (visible and ultraviolet), the **Compton Gamma Ray Observatory**, the **Chandra X-ray Observatory** and the **Spitzer Space Telescope** (infrared). Three were launched by the Space Shuttle. The four never flew all at once — Compton came down three years before Spitzer was launched — but two remain in service in 2026 (Table 39.1).

TABLE 39.1 NASA's four Great Observatories.

Observatory	Band	Telescope	Launch	Orbit	End	Signature results
Hubble	0.115–1.7 µm	2.4 m mirror	24 Apr 1990, STS-31	LEO, 28.5°	Operating	Deep fields; expansion rate; dark energy
Compton (CGRO)	20 keV–30 GeV	4 instruments, 17 t	5 Apr 1991, STS-37	LEO, 28.5°	Deorbited 4 Jun 2000	Gamma-ray bursts are cosmic; blazars
Chandra	0.1–10 keV	1.2 m nested mirrors, 0.5″	23 Jul 1999, STS-93	≈ 64-hour HEO	Operating	Dark matter in Bullet Cluster; black holes
Spitzer	3.6–160 µm	0.85 m, helium-cooled	25 Aug 2003, Delta II	Earth-trailing	30 Jan 2020	TRAPPIST-1 planets; first exoplanet light

39.2.1 Hubble Space Telescope

Hubble is a 2.4 m Ritchey–Chrétien reflector (Chapter 12) with a mass of about 11 t. Space Shuttle Discovery released it on 25 April 1990 at an altitude of about 615 km (Figure 39.3). Within weeks it was clear that the images were blurred: the primary mirror had been polished very precisely to the wrong shape. A 1.3 mm spacing error in the optical test device used during polishing (the null corrector) had made the mirror about 2.2 µm too flat at its edge. This caused **spherical aberration** — light from the edge and the centre of the mirror came to focus at different points.



FIGURE 39.3 Hubble held by Discovery's robotic arm on 25 April 1990, during deployment on mission STS-31, with one solar array unrolled and the second still stowed. Image: NASA/JSC.

Because the error was known exactly, it could be corrected exactly, like prescribing spectacles. The first servicing mission (STS-61, December 1993) installed a new camera with the correction built in and **COSTAR**, small corrective mirrors on deployable arms that fed the other instruments. Astronauts visited Hubble five times between 1993 and 2009, replacing gyroscopes, batteries, solar arrays and every original instrument (Chapters 33 and 40). The last visit installed Wide Field Camera 3 and the Cosmic Origins Spectrograph, which still work alongside two older instruments.

Hubble measured the expansion rate of the Universe, the **Hubble constant**, to about 10 % (72 ± 8 km/s per megaparsec) in 2001. Its observations of distant supernovae helped show in 1998 that the expansion is accelerating — the evidence for **dark energy**. Its **Deep Fields** (1995, 2004), long exposures of apparently empty sky, revealed thousands of galaxies and traced how they grew. It found that almost every large galaxy has a central supermassive black hole and made the first detection of an exoplanet's atmosphere (2001). Its images, such as the "Pillars of Creation" (Figure 39.6), are among the best-known scientific pictures ever made.

Hubble in 2026. On 19 September 2026 Hubble completed its 200 000th orbit, after more than 1.7 million observations. Of the six gyroscopes installed in 2009, two still work, and since June 2024 Hubble has used only one, keeping the other as a spare. This **one-gyro mode** halves the sky it can reach at any moment and initially cut observations by about a tenth, but STScI reported no safe-mode events in the two years that followed. Drag is slowly lowering the orbit, now about 470 km: the Space Telescope Science Institute (STScI) puts the chance of re-entry before 2032 below 10 %, with a median date in late 2033. NASA declined a privately funded Crew Dragon reboost in June 2024, citing risks to the telescope. But it said in 2026 that it would consider a reboost if operating costs fell. STScI has meanwhile been told to plan for budgets about a fifth lower.

39.2.2 Compton Gamma Ray Observatory

At about 17 t, the **Compton Gamma Ray Observatory** (CGRO) was the heaviest astronomical observatory flown up to that time (Figure 39.4). Its four instruments covered photon energies from 20 keV to 30 GeV. The **Burst and Transient Source Experiment** (BATSE) recorded 2 704 **gamma-ray bursts** — brief, intense flashes of gamma rays. It showed that they come equally from all directions rather than concentrating along the Milky Way. Bursts must therefore occur at cosmological distances, which makes them the most powerful explosions known. The **EGRET** telescope showed that the high-energy sky is dominated by **blazars**, galaxies whose central black holes fire jets almost straight at us. After a gyroscope failed in 1999, NASA steered the observatory to a controlled re-entry over the Pacific on 4 June 2000.

39.2.3 Chandra X-ray Observatory

X-rays penetrate a mirror at normal incidence and can be focused only by **grazing incidence**, glancing off a polished surface at about a degree, as a stone skips across water. Chandra's mirror is four nested pairs of nearly cylindrical iridium-coated shells, the largest 1.2 m across, with a 10 m focal length (Chapter 38); its 0.5″ resolution is still the sharpest of any X-ray telescope. Columbia deployed Chandra on 23 July 1999, on the first Shuttle mission commanded by a woman, Eileen Collins (Figure 39.5). An Inertial Upper Stage then lifted it into an orbit that now ranges from 12 650 to 136 200 km above Earth, with a period of 63.5 hours. With its upper stage, Chandra was the heaviest payload ever launched by the Shuttle (22.8 t).

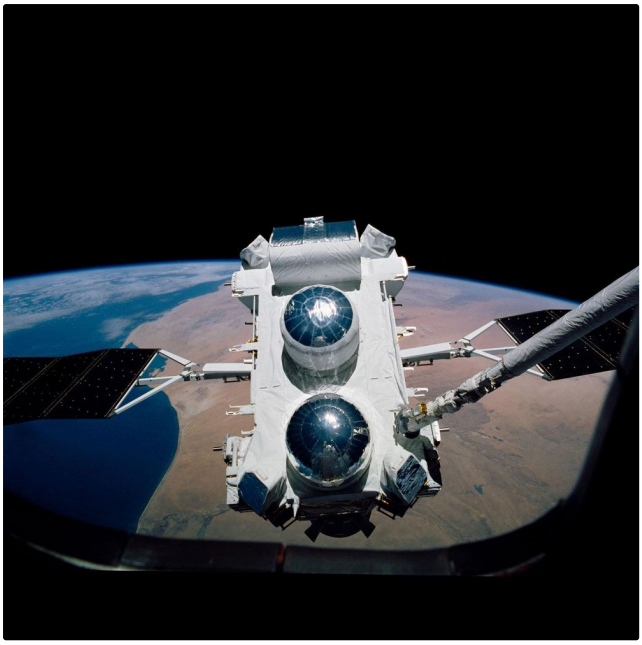


FIGURE 39.4 The Compton Gamma Ray Observatory held by Atlantis's robotic arm during checkout on mission STS-37 in April 1991. The large cylinders are parts of the COMPTEL and EGRET telescopes. Image: NASA/JSC.



FIGURE 39.5 Chandra, with its Inertial Upper Stage, in Columbia's payload bay shortly before being tilted up for release on 23 July 1999. Image: NASA/JSC.

Chandra sees the hot Universe: million-degree gas in galaxy clusters, the remnants of exploded stars and matter falling into black holes. In 2006 its map of hot gas in the colliding **Bullet Cluster** was compared with the mass mapped by gravitational lensing. It showed most of the mass passing through the collision while the gas was slowed — direct evidence for **dark matter**. With JWST in 2023 it detected X-rays from **UHZ1**, a growing black hole seen less than 500 million years after the Big Bang. Figure 39.7 shows how Chandra combines with other observatories. Each wavelength reveals a different part of the same object.



FIGURE 39.6 The "Pillars of Creation" in the Eagle Nebula, re-imaged by Hubble in near-infrared light in 2014. Infrared light passes through much of the dust, revealing stars behind and inside the columns of cold gas. Image: NASA/GSFC.



FIGURE 39.7 The Crab Nebula, the remnant of a supernova seen in 1054, combining radio (Very Large Array), infrared (Spitzer), visible (Hubble), ultraviolet (XMM-Newton) and X-ray (Chandra) data. The X-rays (purple) come from the most energetic particles near the central pulsar. Image: NASA/GSFC.

Chandra has also become a symbol of budget pressure. NASA's requests for 2025 and 2026 proposed winding it down to save about US$ 70 million a year. Congress refused, and the 2026 appropriation (signed in January 2026) gave astrophysics US$ 1.595 billion and Chandra at least US$ 63 million, enough for a "minimum-cost mission". The 2027 request would cut astrophysics by about 65 %, to US$ 552 million, ending Chandra and Fermi. A House committee rejected the cut, but no 2027 budget had been enacted in October 2026. A stopgap continuing resolution held NASA at its 2026 level until 11 December.

39.2.4 Spitzer Space Telescope

Spitzer pioneered a design that JWST later followed: it was **launched warm**. Earlier infrared telescopes such as IRAS (1983) and ISO (1995) sat inside large tanks of liquid helium. Spitzer kept only its instruments in a small cryostat. Its 0.85 m beryllium telescope cooled by radiating to space, helped by vented helium vapour, in an orbit far from the warm Earth (Figure 39.8). The helium lasted until May 2009. Spitzer then worked for ten more years in a "warm mission", using the two shortest-wavelength channels of its camera. These still functioned at the telescope's new temperature of roughly 30 K.

Spitzer made the first detections of light emitted by exoplanets (2005), found a huge, faint ring around Saturn, and in 2017 helped show that the cool dwarf star **TRAPPIST-1** has seven Earth-sized planets. As it drifted away its antenna had to point ever closer to the Sun to send data home, and NASA switched it off on 30 January 2020.



FIGURE 39.8 Spitzer at Cape Canaveral being readied for launch in 2003. The telescope tube sits above the spacecraft bus, shaded by the solar panel on the right. Image: NASA/JPL (NASA).

39.3 The James Webb Space Telescope

JWST was conceived in the 1990s as Hubble's infrared successor, because the ultraviolet and visible light of the first galaxies is stretched by cosmic expansion into the infrared. Built by NASA with the European and Canadian space agencies, it is the largest and most complex scientific spacecraft ever launched (Table 39.2). It took about 25 years to develop and cost about US$ 10 billion, including the first five years of operations.

39.3.1 Launch and Commissioning

JWST's mirror and sunshield were launched folded (Figure 39.9). Its deployment involved about 344 **single-point failures** — items whose failure would have ended the mission. In the two weeks after launch the sunshield unfolded and tensioned, and the mirror wings locked into place. JWST reached L2 on 24 January 2022. Engineers spent three months aligning the 18 segments to within tens of nanometres so that they act as one mirror. The accurate Ariane 5 launch left propellant for about 20 years instead of the 10 planned.

The first images were released on 12 July 2022. The first deep field (Figure 39.10), a 12.5-hour exposure of the galaxy cluster SMACS 0723, reached infrared depths beyond those of Hubble's deepest fields, which had taken weeks. Image quality and background beat requirements. One surprise was **micrometeoroid** damage: an impact in May 2022 slightly distorted one segment, and JWST now limits the time it points towards its direction of orbital motion, where impacts are most energetic.



FIGURE 39.9 The last of JWST's 18 primary-mirror segments being lowered into place by a robotic arm at NASA's Goddard Space Flight Center in February 2016, completing the mirror. Image: NASA/GSFC.

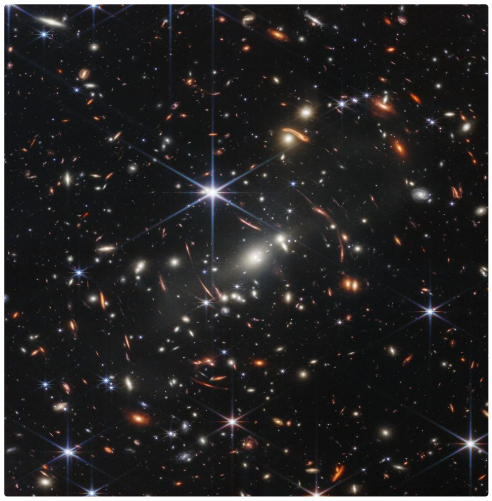


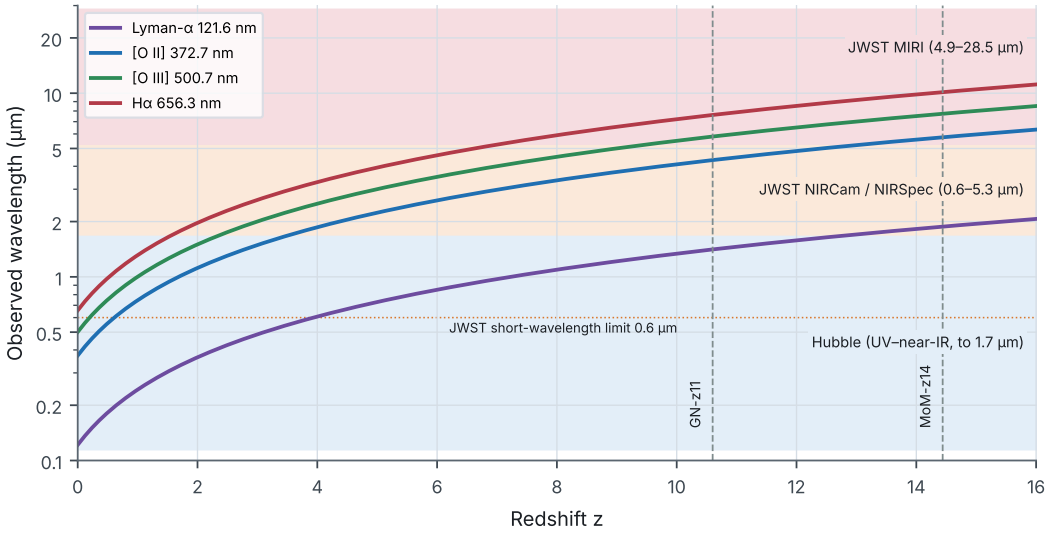
FIGURE 39.10 JWST's first deep field (2022): the cluster SMACS 0723 acts as a gravitational lens, stretching background galaxies into arcs. Image: NASA/STScI (NASA ESA CSA STScI).

TABLE 39.2 The James Webb Space Telescope at a glance.

Parameter	Value
Primary mirror	6.5 m; 18 hexagonal segments of gold-coated beryllium; 25.4 m² collecting area
Wavelength range	0.6–28.5 µm (visible red to mid-infrared)
Instruments	NIRCam (camera), NIRSpec (spectrograph), MIRI (mid-infrared), NIRISS and Fine Guidance Sensor
Sunshield	Five Kapton layers, 21.2 × 14.2 m; Sun side above 350 K, cold side ≈ 40 K
Operating temperature	Telescope ≈ 40 K; MIRI 7 K (cryocooler)
Mass and orbit	About 6.2 t; halo orbit around Sun–Earth L2
Launch	25 December 2021, Ariane 5, Kourou
Lifetime	Propellant for about 20 years

39.3.2 The First Galaxies

Cosmic expansion stretches light from a distant galaxy by a factor $(1+z)$, where z is the **redshift**. Neutral hydrogen absorbs light bluer than the Lyman-α line at 121.6 nm, so a young galaxy vanishes abruptly below $121.6\,(1+z)$ nm. Figure 39.11 shows that for $z > 13$ this **Lyman break** lies beyond 1.7 µm, the red limit of Hubble's cameras. Hubble cannot see such galaxies at all. Hubble's most distant galaxy, GN-z11 (2016), lies at $z = 10.6$.



λ(observed) = λ(emitted) × (1 + z). Shading: wavelength range of the instruments; Earth's atmosphere blocks most light beyond 2.5 µm and below 0.3 µm.

FIGURE 39.11 Observed wavelength of four bright spectral lines as a function of redshift, compared with the ranges of Hubble and JWST's instruments. Above a redshift of about 13, even ultraviolet Lyman-α light arrives at wavelengths longer than Hubble can detect.

JWST has repeatedly broken the distance record, confirming each redshift with a spectrum. In 2024 the JADES survey confirmed **JADES-GS-z14-0** at $z \approx 14.3$ (refined to 14.18 by the ALMA radio telescope). In 2025 the MoM survey measured **MoM-z14** at $z = 14.44$ (published in January 2026), seen just 280 million years after the Big Bang. As of October 2026 no galaxy had been spectroscopically confirmed at a greater distance. The surprise is less the distance than the brightness. JWST finds more luminous galaxies above $z = 10$ than most models predicted. This suggests that early star formation was more efficient than assumed — an active research problem rather than a contradiction of standard cosmology.

JWST has also found black holes earlier and larger than expected. Besides UHZ1, they include a feeding black hole in GN-z11 and many compact, very red **"little red dots"**. These appear to hide black holes of millions of solar masses in dense gas. One of them (CAPERS-LRD-z9, reported in 2025) is seen about 500 million years after the Big Bang. Whether such black holes began as collapsing giant gas clouds or grew unusually fast from the remnants of the first stars is one of JWST's central questions.

39.3.3 Exoplanet Atmospheres

When a planet passes in front of its star, a little starlight filters through its atmosphere. Molecules absorb at characteristic wavelengths, so the planet appears slightly larger at those wavelengths: this **transmission spectrum** reveals the atmosphere's composition. In its first months JWST

made the first clear detection of **carbon dioxide** in an exoplanet atmosphere, that of the hot Saturn WASP-39 b, and found sulphur dioxide made by starlight-driven chemistry. Heat measurements of the rocky planets TRAPPIST-1 b and c suggest little or no thick atmosphere.

K2-18 b, about two and a half times Earth's radius and in the habitable zone of a red dwarf, shows the limits of the method. JWST detected methane and carbon dioxide there in 2023, and in April 2025 a team reported evidence at about three standard deviations for dimethyl sulphide, a gas made on Earth mainly by marine life. Independent reanalyses found that different but reasonable choices of data reduction and atmospheric model remove the signal, and in October 2026 the claim remains unconfirmed. Features from small planets amount to tens of parts per million, close to instrumental limits, so a biosignature claim demands several independent lines of evidence.

39.3.4 JWST in 2026

JWST attracts more demand than any observatory before it: for its fifth cycle, which began on 1 July 2026, astronomers submitted 2 930 proposals requesting about twelve times the time available. It has confirmed that Hubble's measurements of the variable stars calibrating the distance ladder were not biased by crowding. Yet calibrations still give Hubble constants of about 70–73 km/s/Mpc, against 67–68 km/s/Mpc from the cosmic microwave background — the unexplained **Hubble tension**. Its budget is under pressure too: the 2027 request proposed reduced operating funds for JWST and Hubble.

39.4 Reading the Distant Universe: Redshift, Time and Distance

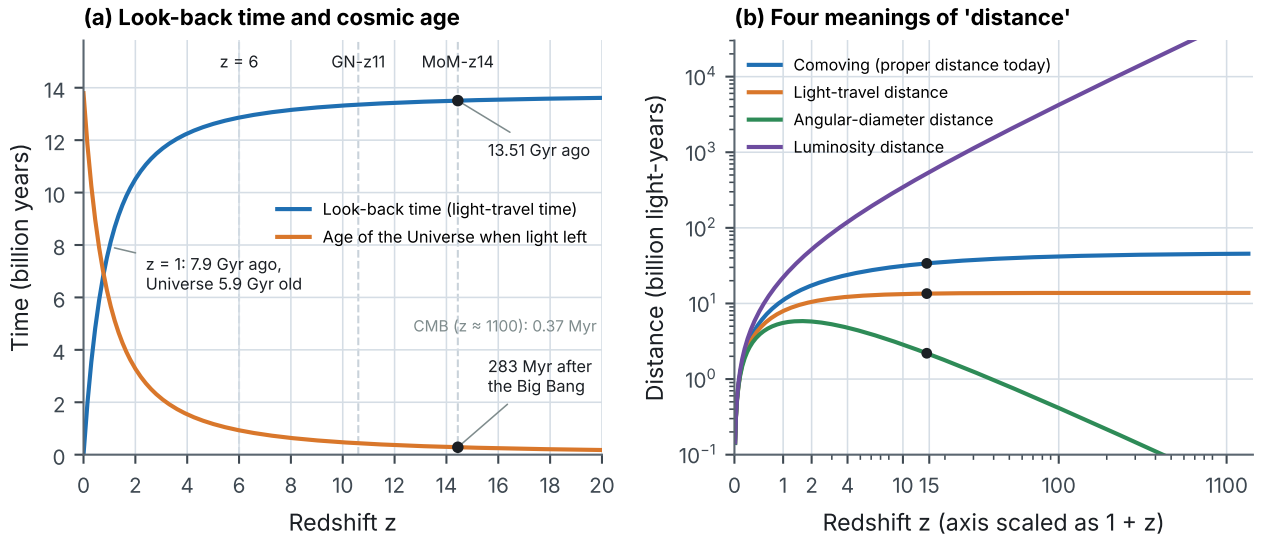
Looking far away means looking back in time, but in an expanding Universe "how far" has several meanings. The standard model is **flat ΛCDM**: a Universe dominated today by dark energy, Λ, and cold dark matter. With Hubble constant H_0 = 67.7 km/s/Mpc and matter density Ω_m = 0.31, it gives a present age of 13.79 billion years. The age of the Universe when light was emitted at redshift z is

$$t(z) = \int_z^\infty \frac{dz'}{(1+z')\,H(z')}, \qquad H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda},$$

where $\Omega_r \approx 9 \times 10^{-5}$ is the small contribution of radiation and $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$. The **look-back time** is the present age minus $t(z)$ (Figure 39.12, Table 39.3). Four distances are in common use:

- **Light-travel distance**: look-back time × speed of light — the figure press releases quote, though no ruler could measure it.
- **Comoving distance**: the distance to the object *today*, if the expansion could be frozen and a ruler laid out. It exceeds the light-travel distance because space expanded while the light was in transit; the edge of the observable Universe is about 46 billion light-years away.
- **Angular-diameter distance**: relates an object's true size to the angle it covers. It peaks near $z \approx 1.6$ and then *decreases*: very distant galaxies look larger than galaxies of the same size at intermediate distance, because we see them as they were when they were closer to us.

- **Luminosity distance**: relates an object's power to the flux received, as used with "standard candles" such as type Ia supernovae. It exceeds the comoving distance by a factor $(1+z)$, because photons lose energy and arrive less often.



Flat ΛCDM: H_0 = 67.7 km/s/Mpc, Ωm = 0.31, Ωr = 9.1 × 10^{-5}, ΩΛ = 1 − Ωm − Ωr. Present age 13.79 Gyr. Dots: MoM-z14.

FIGURE 39.12 (a) Look-back time and age of the Universe when the light was emitted, as functions of redshift. (b) Four measures of distance. Light from MoM-z14 travelled for 13.5 billion years, but the galaxy is now about 34 billion light-years away.

EXAMPLE: LOOK-BACK TIME TO REDSHIFT 14

A galaxy's spectrum shows the Lyman break at 1.824 µm. How long ago did its light leave, and how old was the Universe then?

1. **Redshift.** $1+z = 1.824/0.1216 = 15.0$, so $z = 14.0$. The break lies beyond Hubble's 1.7 µm limit; only JWST can measure it.
2. **Quick estimate.** At this redshift matter dominates, $H(z) \approx H_0\sqrt{\Omega_m}(1+z)^{3/2}$, and the integral gives $t \approx \frac{2}{3}\,(1/H_0)/[\sqrt{\Omega_m}(1+z)^{3/2}]$. With $1/H_0$ = 14.44 Gyr: $t \approx \frac{2}{3} \times 14.44/(0.557 \times 58.1)$ = **0.298 Gyr**.
3. **Exact integral.** Numerical integration including dark energy and radiation gives t = **296 million years**, within 1 % of the estimate — dark energy was negligible then.
4. **Look-back time.** 13.79 − 0.30 = **13.50 billion years**: the light has travelled for 97.9 % of cosmic history.

TABLE 39.3 Time and distance in flat ΛCDM (H_0 = 67.7 km/s/Mpc, Ω_m = 0.31); 1 Gly = 10^9 light-years.

Redshift z	Example	Look-back time (Gyr)	Age then (Myr)	Comoving distance (Gly)	Angular-diameter distance (Gly)
1	Star-formation peak	7.94	5 860	11.1	5.54
2	Quasar era	10.51	3 280	17.3	5.77
6	Reionisation ends	12.86	930	27.5	3.93
10.6	GN-z11	13.36	436	31.9	2.75
14.44	MoM-z14	13.51	283	33.9	2.19
1 100	CMB	13.79	0.37	45.3	0.041

EXAMPLE: HOW FAR, AND HOW BIG?

For the same galaxy at $z = 14.0$, compare the distances and decide whether JWST can resolve it.

1. **Light-travel distance**: 13.5 billion light-years (Gly), the number usually quoted.
2. **Comoving distance today**: integrating $c/H(z)$ from 0 to 14 gives 10 330 Mpc = **33.7 Gly**, 2.5 times the light-travel distance.
3. **Distance when the light left**: comoving distance ÷ $(1+z)$ = 33.7/15.0 = **2.25 Gly**, only a sixth of the light-travel distance. In a flat universe this equals the angular-diameter distance.
4. **Angular size**: a galaxy 500 parsecs (1 630 light-years) across subtends $500\ \mathrm{pc}/688.6\ \mathrm{Mpc} = 7.3 \times 10^{-7}$ rad = **0.15″**.
5. **Resolution**: JWST's diffraction limit at 2 µm is $1.22 \times 2 \times 10^{-6}/6.5$ = 3.75×10^{-7} rad = **0.077″**, half the galaxy's size, so JWST resolves it into about two resolution elements across. Hubble at 1.6 µm (0.17″) could not, even if the light were in its band.
6. **Luminosity distance**: $(1+z)$ × comoving distance = **505 Gly**, so the flux is $15^2 = 225$ times lower than an inverse-square estimate using the comoving distance.

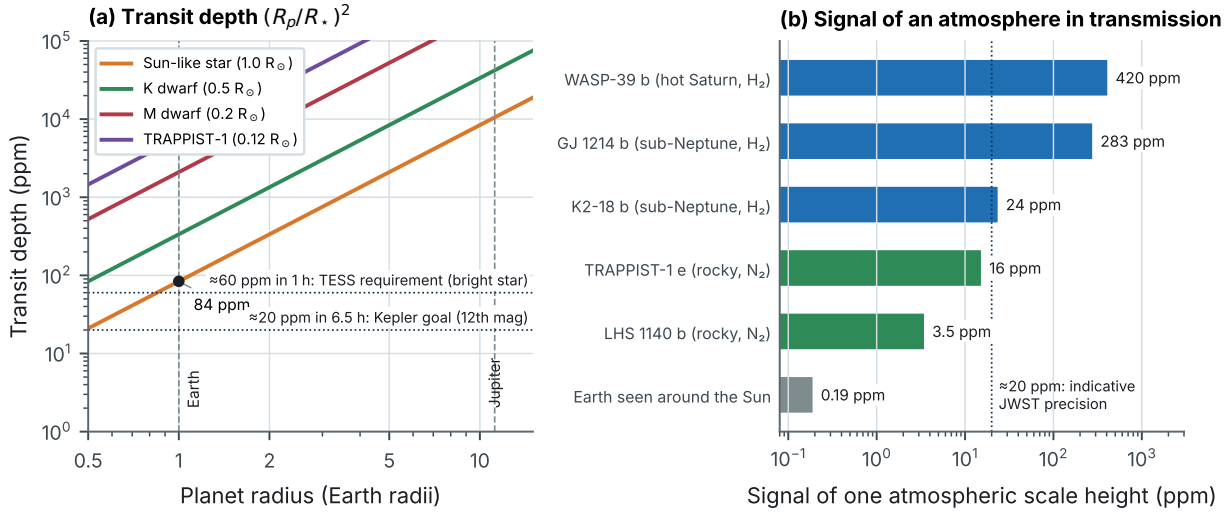
39.5 Surveys and Exoplanet Hunters

Survey missions observe large parts of the sky systematically to build catalogues, measure the shape of the Universe or find rare events. Their power depends less on aperture than on **étendue**, the product of collecting area and field of view, which sets how fast they map the sky (Figure 39.18 b).

39.5.1 Finding Planets: Kepler and TESS

When a planet of radius R_p crosses a star of radius $R_\star$, the star dims by the **transit depth** $\delta = (R_p/R_\star)^2$ (Figure 39.13 a). This is 84 parts per million (ppm) for Earth crossing the Sun, and about 1 % for Jupiter. Only a fraction $R_\star/a$ of systems, for a planet at distance a, are aligned so that transits occur — 0.47 % for an Earth–Sun analogue. So a transit survey must monitor tens of thousands of stars or more.

NASA's **Kepler** (2009–18) used a 0.95 m Schmidt telescope and a 95-megapixel camera covering 115 square degrees (Figure 39.15) to measure about 150 000 stars in Cygnus and Lyra every 30 minutes for four years. When a second reaction wheel failed in 2013, engineers balanced the spacecraft against sunlight pressure, and the **K2** mission observed fields along the ecliptic until the propellant ran out in October 2018. Kepler showed that planets are the rule rather than the exception. It showed that the commonest planets in the Galaxy are between Earth and Neptune in size — a type absent from the Solar System. It also showed that a substantial fraction of Sun-like stars (estimates range from about one in ten to one in two) have an Earth-sized planet in the habitable zone.



Scale height H = kT/(μ m_H g), T = equilibrium temperature; signal = 2H·Rp/R*²; strong bands span 2–5 H. Data: NASA Exoplanet Archive; μ = 2.3 (H_2) or 28–29 (N_2-rich) assumed.

FIGURE 39.13 (a) Transit depth versus planet radius for four types of star, with the photometric precision of Kepler and TESS. (b) Signal of one atmospheric scale height in transmission for selected planets: hydrogen-rich planets give tens to hundreds of ppm, rocky planets with nitrogen-rich air from a fraction of a ppm (Earth around the Sun) to about 15 ppm (TRAPPIST-1 e around a tiny star).

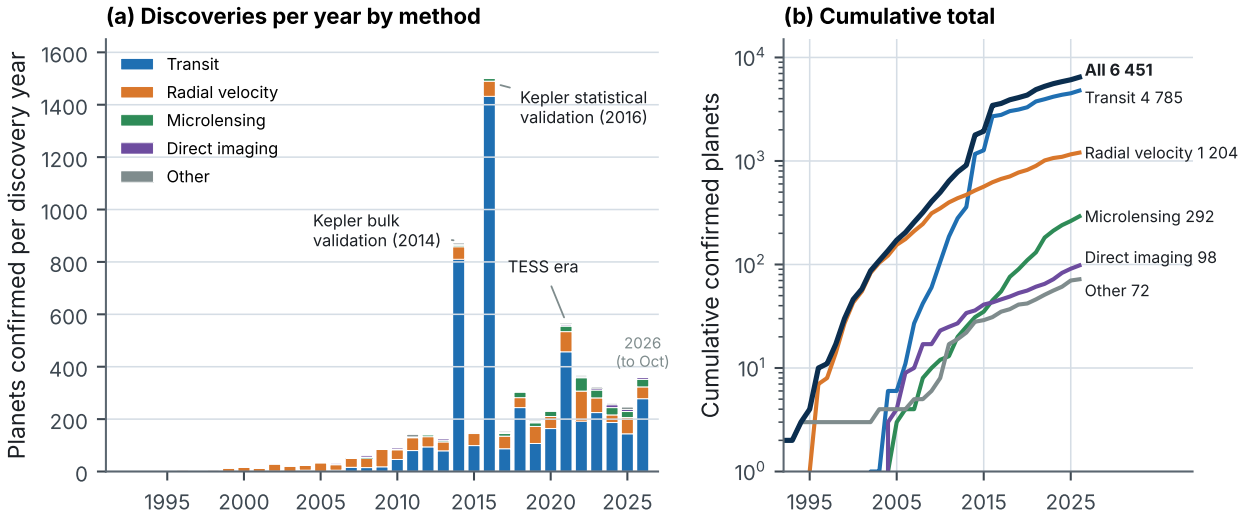
The **Transiting Exoplanet Survey Satellite (TESS)**, launched in 2018, has four 10.5 cm cameras covering a 24° × 96° strip of sky, which it observes for 27 days before moving on. It has covered more than 98 % of the sky, concentrating on bright, nearby stars whose planets can be weighed from the ground and studied by JWST. By October 2026 it had flagged 8 151 "TESS Objects of Interest" — possible planets, some later shown to be false alarms. Of its discoveries, 1 012 were confirmed planets. It continues in its third extended mission. ESA's **CHEOPS** (2019), a 32 cm telescope, measures the radii of known planets precisely.

EXAMPLE: CAN TESS FIND AN EARTH TWIN?

An Earth-sized planet orbits a Sun-like star (1.0 $R_\odot$, 1.0 $M_\odot$) once a year. For a bright star TESS reaches about 60 ppm of noise in one hour.

1. **Depth**: $\delta = (6\,371/695\,700)^2 = 8.4 \times 10^{-5}$ = **84 ppm**.
2. **Duration** of a central transit: $T \approx (P/\pi)\,(R_\star/a)$ = (8 766 h/π) × (695 700/149.6 million) = **13.0 hours**.
3. **Noise over one transit**: 60 ppm/√13 = 16.6 ppm, giving a signal-to-noise ratio of 84/16.6 = **5.0**.
4. **Detection** conventionally needs a ratio of about 7.1, i.e. $(7.1/5.0)^2 \approx$ **2 transits** — a continuous baseline of more than a year. TESS watches most stars for only 27 days, so it essentially cannot find Earth twins, and only 0.47 % of such systems transit at all. That is the job of PLATO, which will stare at the same fields for two years or more.
5. **Around a small red dwarf** (0.2 $R_\odot$), the same planet in the habitable zone (about 0.07 AU) orbits in 14 days with a depth of **2 100 ppm**. Even with three times the noise from the fainter star, one 1.5-hour transit gives a ratio of about 14, and TESS sees two transits per sector. This is why most known small temperate planets — and those JWST can study — orbit red dwarfs.

By October 2026 the NASA Exoplanet Archive listed **6 451 confirmed planets** (Figure 39.14), with more than 8 000 candidates awaiting confirmation. Three-quarters (4 785) were found by transits, and over half (3 337) by Kepler and K2. Table 39.4 summarises the main space missions.



Data: NASA Exoplanet Archive, Planetary Systems Composite table, queried 19 Oct 2026 (6 451 planets). 'Other' = timing, astrometry, brightness modulation, disk kinematics.

FIGURE 39.14 Exoplanet discoveries by year and method (NASA Exoplanet Archive, October 2026). The peaks in 2014 and 2016 are statistical validations of large batches of Kepler candidates.

TABLE 39.4 Space missions for finding and characterising exoplanets.

Mission	Years	Method	Telescope / field	Role and status (Oct 2026)
Kepler / K2 (NASA)	2009–18	Transit	0.95 m, 115 deg²	Planet statistics; 3 337 planets
TESS (NASA)	2018–	Transit	4 × 10.5 cm, 2 304 deg²	Bright nearby stars; 1 012 planets
JWST (NASA/ESA/CSA)	2021–	Spectroscopy, imaging	6.5 m	Atmospheres of known planets
Roman (NASA)	2026–	Microlensing, coronagraph	2.4 m, 0.28 deg²	Cold planets; >1 000 expected
PLATO (ESA)	2027–	Transit	26 × 12 cm	Earth analogues around Sun-like stars
Ariel (ESA)	2029–	Spectroscopy	1.1 × 0.7 m	Atmospheres of about 1 000 planets
HWO (NASA)	2040s	Direct imaging	≈ 6 m	Search for life on ~25 Earth-like planets

39.5.2 Gaia: Mapping the Galaxy

ESA's **Gaia** (2013–25) measured the positions, distances and motions of nearly two billion stars from L2. Two telescopes 106.5° apart scanned the sky as the spacecraft rotated, imaging onto nearly a billion pixels. As a result, angles between stars were measured to ten to a few tens of micro-arcseconds — roughly the size of a coin on the Moon seen from Earth. Its **parallaxes**, the annual shifts of stars caused by Earth's orbit, give distances to over a billion stars. Its data also revealed the debris of a dwarf galaxy that merged with the Milky Way about 10 billion years ago. Science operations ended on 15 January 2025; the fourth data release, from 5.5 years of data and more than 500 TB, is scheduled for December 2026.

39.5.3 Mapping the Dark Universe: Euclid, SPHEREx, Roman and Xuntian

Euclid (ESA, July 2023) is a 1.2 m telescope with a 0.53-square-degree field — over a hundred times that of Hubble's main camera. It is mapping a third of the sky in six years to measure the shapes of more than a billion galaxies. Their tiny, coherent distortions by intervening matter (**weak gravitational lensing**) and the clustering of galaxies reveal how dark matter is distributed and how dark energy has driven the expansion. Its first quick release (March 2025) covered 63 square degrees and 26 million galaxies. The first major data release (DR1) begins in November 2026 with images and catalogues of about 1 900 square degrees. The weak-lensing and clustering products needed for cosmology follow in mid-2027. NASA's **SPHEREx** (March 2025) splits light into 102 colours from 0.75 to 5 μm with a 20 cm telescope. It completed the first of four planned all-sky spectral maps in December 2025. It seeks imprints of **cosmic inflation** — the extremely rapid expansion thought to have occurred in the first instant after the Big Bang — and surveys interstellar ices.

The **Nancy Grace Roman Space Telescope** (NASA, about US$ 4 billion) pairs a 2.4 m mirror, donated by the US National Reconnaissance Office, with a 300-megapixel camera. The camera's field is about 200 times that of Hubble's infrared camera. Launched on a Falcon Heavy on 30 August 2026, ahead of schedule (Figure 39.16), it was being commissioned on its way to L2 in October 2026, with first science images expected in early 2027. Over five years it will map dark energy. It will find more than a thousand planets towards the Galactic centre by **gravitational microlensing** (the brief brightening of a background star as a planet passes in front of it). It will also test a coronagraph required to detect planets 10^{7} times fainter than their star (ground tests reached better than 5×10^{-8}). This is a step towards the 10^{10} needed for Earth twins.



FIGURE 39.15 The Kepler spacecraft in a clean room at Ball Aerospace in Boulder, Colorado, in 2008. The white dust cover on the sunshade was jettisoned after launch. Image: NASA/JPL (NASA/JPL-Caltech/Ball Aerospace).



FIGURE 39.16 The Nancy Grace Roman Space Telescope being encapsulated in its Falcon Heavy payload fairing at Kennedy Space Center on 21 August 2026, nine days before launch. Image: NASA/KSC (NASA/Sydney Rohde (Rocz)).

China's 2 m **Xuntian** survey telescope (CSST, 1.1-square-degree field) is to be launched in 2027 into the orbit of the Tiangong station, where it can dock for servicing. ESA's **PLATO**, with 26 small cameras, is due for launch on Ariane 6 in early 2027 (ESA's mission page gives April). Its aim is to find Earth-sized planets in the habitable zones of Sun-like stars. **Ariel** (2029) will measure the atmospheres of about a thousand exoplanets.

39.6 High-Energy and Time-Domain Astronomy

X-ray and gamma-ray observatories study the most violent events in the Universe, many lasting seconds to days. So high-energy astronomy is a **time-domain** discipline: wide-field monitors detect a burst and narrow-field telescopes swing onto it within minutes.

- **Fermi** (NASA, 2008) carries the Large Area Telescope (20 MeV to over 300 GeV), which surveys the whole gamma-ray sky every three hours and has catalogued about 7 000 sources. It combines this with a burst monitor (8 keV–40 MeV). On 17 August 2017 it detected a short gamma-ray burst 1.7 seconds after LIGO and Virgo recorded gravitational waves from merging neutron stars, opening **multi-messenger astronomy**.
- **Swift** (NASA, 2004) could turn to a new burst in about a minute and observe it with X-ray and ultraviolet/optical telescopes; it observed more than 2 000 gamma-ray bursts. The high solar activity of 2024–25 made its orbit decay much faster than predicted. NASA paid Katalyst Space US$ 30 million for a robotic spacecraft to grab Swift and raise its orbit. Launched on a Pegasus XL on 3 July 2026, it suffered attitude-control problems, and the attempt was abandoned on 19 August. Below 275 km drag prevented steady pointing: Swift made its last observation on 20 October 2026, NASA ended the mission on 23 October, and re-entry is expected within weeks.
- **NuSTAR** (NASA, 2012) was the first telescope to focus hard X-rays (3–79 keV), using a 10 m mast unfolded in orbit for the long focal length grazing-incidence optics need. **IXPE** (NASA–Italy, 2021) measures X-ray **polarisation**, revealing the geometry of magnetic fields in pulsars and black-hole jets.
- **XRISM** (JAXA with NASA and ESA, 2023) carries **Resolve**, a microcalorimeter. It measures each X-ray photon's energy from the tiny temperature rise it causes in a detector at 0.05 K. Its spectral resolution is 20–30 times finer than that of X-ray CCDs. Its protective gate valve has failed to open, blocking X-rays below about 1.7 keV, but it works superbly at higher energies and operations have been extended to 2030.
- **eROSITA**, the German telescope on Russia's Spektr-RG (2019), completed four all-sky X-ray surveys before being idled in February 2022 after Russia's invasion of Ukraine; its July 2026 release lists about two million sources.
- **Einstein Probe** (China with ESA, Germany and France, January 2024) uses "lobster-eye" optics, modelled on crustacean eyes, to watch 3 600 square degrees at once in soft X-rays. It has found many new transients. The Chinese–French **SVOM** (June 2024) detects gamma-ray bursts and follows up its alerts. ESA's **XMM-Newton** (1999) remains Europe's X-ray spectroscopy workhorse.

39.7 Microwaves, Radio and Gravitational Waves

The **cosmic microwave background (CMB)** is light released 370 000 years after the Big Bang, when the Universe became transparent, now stretched 1 100-fold into microwaves. NASA's **COBE** (1989) showed that it has a perfect blackbody spectrum at 2.725 K. It also detected temperature variations of a few parts in 100 000 — the seeds of all later structure. This work was recognised by the 2006 Nobel Prize in Physics. **WMAP** (2001–10) mapped those variations over the whole sky from L2, and ESA's **Planck** (2009–13) mapped them in nine frequency bands with about three times better resolution. Planck's final results (2018) gave an age of 13.8 billion years, a composition of about 5 % ordinary matter, 27 % dark matter and 68 % dark energy, and $H_0 = 67.4 \pm 0.5$ km/s/Mpc.

In radio astronomy space extends the size of an interferometer. Russia's **RadioAstron** (2011–19) linked a 10 m orbiting antenna, reaching about 350 000 km from Earth, with ground radio telescopes. It achieved resolutions of about ten micro-arcseconds.

The next frontier is not light at all. The **Laser Interferometer Space Antenna (LISA)** was adopted by ESA in January 2024 with NASA participation and is planned for launch on Ariane 6 in 2035. It will fly three spacecraft in a triangle 2.5 million km on a side and use lasers to measure picometre changes in their separations. It will detect low-frequency **gravitational waves** from merging supermassive black holes and compact binary stars, which ground-based detectors such as LIGO cannot hear.

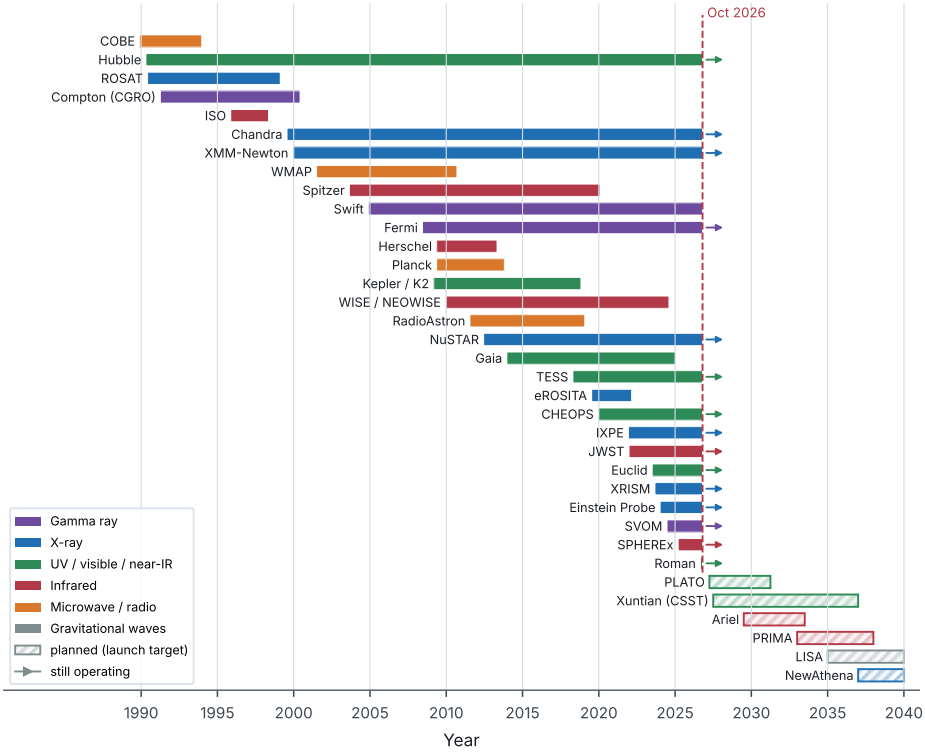
39.8 The Fleet in 2026 and the Road Ahead

Figure 39.17 shows how missions have succeeded one another since 1990, and Table 39.5 lists the main observatories operating or approved in October 2026. Missions outlive their plans — Hubble was designed for 15 years and Chandra for five. Europe, China and Japan now lead many missions. Apertures grow slowly, limited by fairings and cost (Figure 39.18 a), so progress comes largely from wider fields, better detectors and colder telescopes.

TABLE 39.5 Major space observatories operating or approved in October 2026 (launch years of future missions are targets).

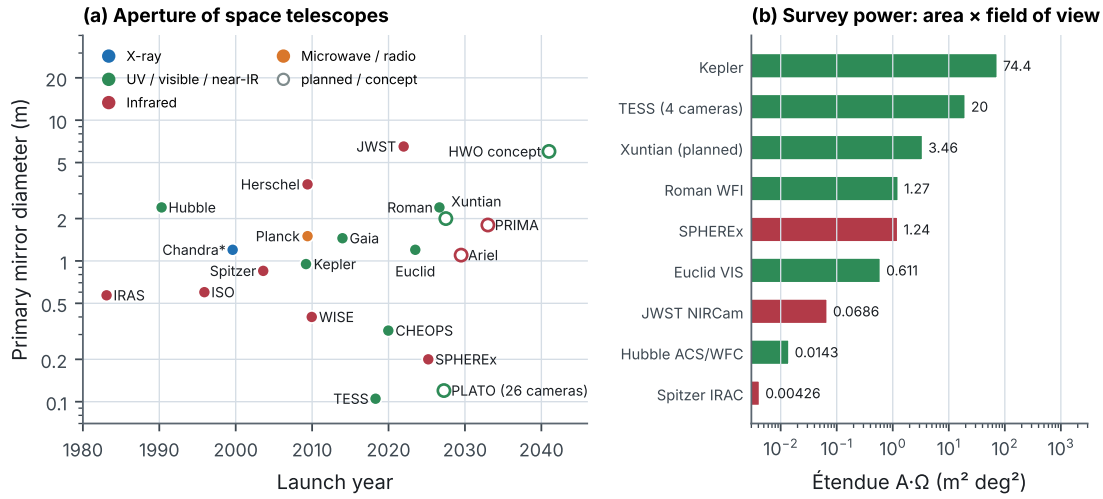
Mission	Lead	Band	Aperture	Orbit	Launch	Status
Hubble	NASA/ESA	0.115–1.7 µm	2.4 m	LEO 470 km	1990	One-gyro mode
Chandra	NASA	0.1–10 keV	1.2 m	HEO	1999	Minimum-cost
Swift	NASA	γ, X-ray, UV	0.3 m	LEO 265 km	2004	Ended Oct 2026
Fermi	NASA	8 keV–300 GeV	Detectors	LEO 490 km	2008	Budget threat
TESS	NASA	0.6–1.0 µm	4 × 10.5 cm	HEO	2018	Extended
JWST	NASA/ESA/CSA	0.6–28.5 µm	6.5 m	L2	2021	Cycle 5
Euclid	ESA	0.55–2.0 µm	1.2 m	L2	2023	Surveying
XRISM	JAXA/NASA	1.7–12 keV	2 × 0.45 m	LEO 525 km	2023	Operating
Einstein Probe	CAS/ESA	0.3–10 keV	Lobster-eye	LEO 600 km	2024	Operating
SPHEREx	NASA	0.75–5.0 µm	0.2 m	LEO 650 km	2025	Surveying

Mission	Lead	Band	Aperture	Orbit	Launch	Status
Roman	NASA	0.48–2.3 µm	2.4 m	L2	2026	Commissioning
PLATO	ESA	0.5–1.0 µm	26 × 0.12 m	L2	2027	Final testing
Xuntian	China	0.25–1.0 µm	2 m	LEO	2027	Building
Ariel	ESA	0.5–7.8 µm	1.1 × 0.7 m	L2	2029	Building
PRIMA	NASA	24–235 µm	1.8 m	L2	2033	Selected
LISA	ESA/NASA	Grav. waves	3 × 0.3 m	Solar orbit	2035	Adopted
NewAthena	ESA	0.2–12 keV	≈ 2.5 m	L-point	2037	Study
HWO	NASA	UV–near-IR	≈ 6 m	L2	2040s	Concept



Bars run from launch to end of science operations. Planned bars start at the launch target (Oct 2026) and show nominal mission length; Swift ended science on 20 Oct 2026 ahead of re-entry.

FIGURE 39.17 Timeline of major space observatories, 1990–2040. Bars show operating life from launch; hatched bars are planned missions starting at their target launch dates.



*Chandra: outer diameter of nested grazing-incidence mirrors. HWO: ~6 m concept, 2040s. Étendue uses geometric aperture area (JWST 25.4 m²) and the field of the main survey camera(s).

FIGURE 39.18 (a) Primary-mirror diameter of space telescopes against launch year. (b) Survey power (étendue): collecting area multiplied by the field of view of the main camera. Wide-field survey telescopes map the sky thousands of times faster than Hubble or JWST.

39.8.1 New Missions and the Habitable Worlds Observatory

In September 2026 NASA selected **PRIMA**, a cryogenically cooled 1.8 m far-infrared telescope, as the first mission in a new **Probe** class with a cost cap of US$ 1.2 billion. Due for launch in 2033, it will study how galaxies, stars and planets form, filling the gap left by Spitzer and Herschel. The UVEX ultraviolet explorer (about 2030) and Israel's ULTRASAT (2027) will survey the ultraviolet sky, and ESA's **NewAthena** X-ray observatory is planned for adoption in 2027 and launch in the late 2030s.

The 2021 US astronomy decadal survey recommended as NASA's next great observatory a telescope of about 6 m to image Earth-like planets around Sun-like stars and search their spectra for oxygen, water and methane. This **Habitable Worlds Observatory (HWO)** needs a coronagraph suppressing starlight by 10^{10}. An Earth seen from 30 light-years is ten billion times fainter than its star and only 0.1″ from it. It also needs a wavefront stable to picometres. NASA aims for launch in the early 2040s, but funding is uncertain: Congress provided about US$ 150 million for 2026, while the 2027 request proposed almost nothing.

Astronomers thus have more capability in space than ever, yet budget requests propose ending productive missions that cost tens of millions a year to run after costing billions to build.

AI IN SPACE ASTRONOMY

Machine learning classifies **transients**: ground surveys such as the Vera C. Rubin Observatory, which released its first images in 2025, are expected to issue millions of alerts per night. Classifiers in "alert brokers" decide within minutes which supernovae, tidal-disruption events or X-ray flares deserve follow-up by satellites such as Einstein Probe. In **exoplanet vetting**, neural networks trained on Kepler light curves separate true transits from eclipsing binaries and instrumental artefacts. One found the eighth planet of Kepler-90 in 2017, and NASA's ExoMiner validated 301 new Kepler planets in 2021. **Galaxy morphology** is classified by deep networks trained on volunteers' labels from the Galaxy Zoo project, now applied to Euclid images. All such outputs must be validated, because a subtle bias can mimic a discovery (Chapter 51).

KEY TAKEAWAYS

- Space telescopes gain wavelengths the atmosphere blocks, diffraction-limited sharpness and photometric stability, at hundreds of times the ground's cost per square metre.
- Of NASA's Great Observatories, Hubble (one-gyro mode, median re-entry late 2033) and Chandra (minimum-cost operations) still worked in October 2026; Compton and Spitzer ended in 2000 and 2020.
- JWST has seen galaxies to $z = 14.44$, 280 million years after the Big Bang, found surprisingly early black holes and measured exoplanet atmospheres; the K2-18 b biosignature claim is unconfirmed.
- Light from $z = 14$ left 13.5 billion years ago from a source now 34 billion light-years away, which was 2.25 billion light-years away when the light left: cosmic distances must be defined carefully.
- A transiting Earth dims a Sun-like star by 84 ppm for 13 hours. Of 6 451 confirmed exoplanets (October 2026), three-quarters were found by transits; PLATO will hunt true Earth analogues.
- Survey telescopes gain power from wide fields; high-energy missions form a rapid-response network.
- PLATO and Xuntian (2027), Ariel (2029), PRIMA (2033), LISA (2035), NewAthena and the Habitable Worlds Observatory lie ahead, but budget threats to operating missions are serious.

REVIEW QUESTIONS

1. Give three reasons for putting a telescope in space, with an example of each.
2. What caused Hubble's spherical aberration, and why could it be corrected completely?
3. Compare LEO, HEO and an L2 halo orbit for an infrared telescope and for an X-ray telescope.
4. Why must X-ray telescopes use grazing-incidence mirrors, and with what consequences?
5. Why can JWST, but not Hubble, see galaxies beyond $z \approx 13$?
6. Using Table 39.3, give the look-back time, age at emission and comoving distance for $z = 6$. Why does the comoving distance exceed the light-travel distance?
7. Find the transit depth and duration for a 3.9 Earth-radius planet in a 30-day orbit around a Sun-like star.
8. Why is the K2-18 b dimethyl-sulphide claim unconfirmed, and what would establish a biosignature?
9. Define étendue, and explain why SPHEREx, with a 20 cm telescope, maps the sky faster than JWST.
10. Discuss the case for and against funding old observatories such as Hubble and Chandra when new missions need money.